SigmaStar

Camera calibration instructions

REVISION HISTORY

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1.8	• Added EIS function	09/30/2024
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1. OVERVIEW

1.1. Overview

This article describes how to use the CVTool tool for multi-camera calibration and stitching functions, as well as single-camera calibration functions. According to the calibration images captured by the user from multiple cameras or single cameras, the calibration parameters required by the chip are calculated.

2. CALIBRATION TOOL ENVIRONMENT INSTALLATION

Camera calibration uses the CVTool tool of SigmaStar Company, which depends on the Visual Studio runtime library. Make sure that the runtime library of VS 2017 is installed before running. You can download vc2017setup.exe to install it by yourself. If the tool can be opened normally, skip this step. Otherwise, open the installation wizard and check Visual C++ Redistributable Package 2017, uncheck the other options, and then click Next to install, as shown in Figure 1:

Figure 1: Visual Studio runtime library installation1

3. CAMERA CALIBRATION

3.1. Preparation before camera calibration

3.1.1 AE/AWB synchronization

In most splicing products today, multiple ISPs are independent of each other, which leads to differences in brightness, color and contrast between different sensors, especially in scenes with uneven brightness and dark. Therefore, before camera calibration, it is necessary to do a good job of AE and AWB synchronization.

- AE Synchronization
  Auto Exposure (AE) is a camera that automatically adjusts to overexposure or underexposure depending on the intensity of the light, so that the image area is not too bright or too dark. Due to the different placement Angle and distance between adjacent cameras and the photographed object, the photographed area is also different, resulting in different exposure of each lens. Therefore, before stitching, we must synchronize the multiple lenses with AE to avoid the discontinuity of light and dark parts at the stitching place. The stitching difference before and after AE synchronization is shown in FIG. 2, where the right subfigure is the picture display after AE synchronization. Refer to the Settings of MI_ISP_SYNC3A_e in "MI_ISP_API_V3.doc" for SigmaStar AE synchronization.

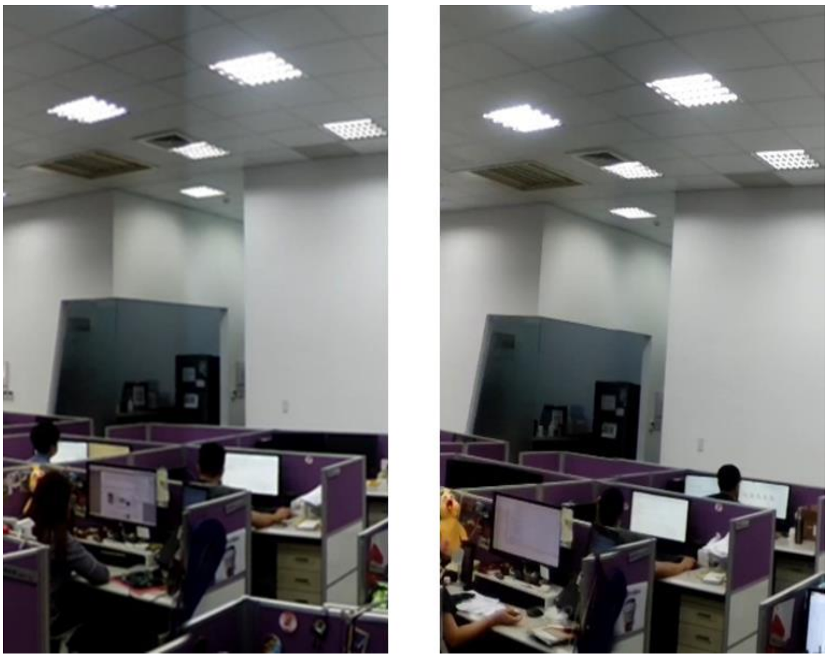

FIG. Splicing difference diagram before and after AE synchronization2

- AWB (Auto White Balance) synchronization

  Under different color temperature light sources, photos taken by the same white object may appear color cast, such as red in low temperature and blue in high temperature. Due to the assembly errors between cameras of the same model, the AWB parameters of each camera will also be slightly different. In order to ensure the quality of the stitching, it is necessary to synchronize the AWB of multiple cameras before stitching. AWB synchronization method is the same as AE, refer to MI_ISP_SYNC3A_e Settings in "MI_ISP_API_V3.doc".

3.1.2 Lens Shading Correction

Due to the process and other optical problems of the lens, the image captured by the lens appears brighter at the center and darker at the edge. In the scene of multi-camera Mosaic, there will be some differences in the brightness of the overlapping area. In order to ensure the fusion effect at the Mosaic, it is recommended to perform lens shading correction on each camera first.

3.1.3 Check, debug and import the Image Quality IQ(Image Quality) bin suitable for calibration

Before calibration, it is necessary to check and debug the image quality IQ suitable for calibration, and import the IQ bin to the board. The adjusted IQ should be able to detect corners as accurately as possible to improve the image quality used for camera calibration. The before-and-after effect comparison of IQ used for camera calibration is shown in Figure 3 and Figure 4. The reference debugging scheme of IQ suitable for camera calibration is as follows:

1. Increase the black and white contrast: The YUV Gamma is drawn into a small s shape to make the corners more visible;

2. WDR_LCE module: By fine-tuning the relevant parameters, make the picture black and white (Num, Filter is increased; Black darkening/white brightening intensity is increased), which can open the difference between the pixel intensity of the corner and the surrounding pixels, so that the corner is highlighted;

3. YEE module: try to go full low and medium frequency (ML) or high frequency (H), the effect on corner detection is not obvious; Try full walk with direction (D) or without direction (UD), the effect on corner detection is not obvious; Increase EdgeGain, Edgekillut, OverShootGain to improve the clarity of the checkerboard;

4. Sharpen the white edge not too strong;

5. Increase the brightness and increase the exposure time: if there is a strong noise point in the picture, it will have a certain impact on the corner detection. You can increase the image brightness and increase the exposure time to reduce the noise points.

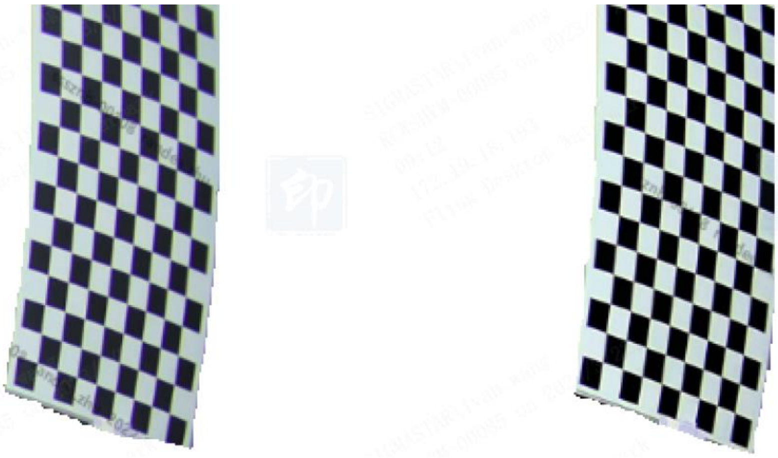

FIG. 3 Comparison of calibration map effects before and after IQ adjustment3

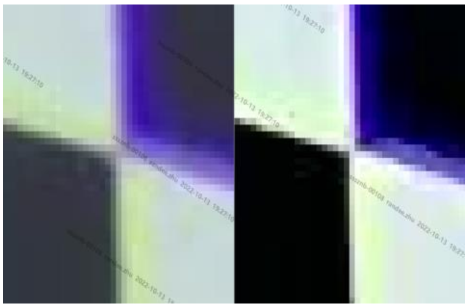

Figure 4 Comparison of diagonal-checkerboard Hague effects before and after IQ tuning4

As shown in Figure 3, the left picture is the calibration effect before debugging the IQ according to the above scheme, and the right picture is the calibration effect after debugging the IQ. Compared with the left picture, the checkerboard is clearly clearer and the black and white contrast is larger in the right picture. Figure 3 Zoom in so that the human eye can clearly see the pixel-wise corner areas, as shown in Figure 4. The left image is the enlarged image of the left image of Figure 3, and the right image is the enlarged image of the right image of Figure 3. Compared with the left image, the transition area between the diagonal black checkerboard squares in the right image is smaller, the pixel intensity of the corner points is significantly different from that of the surrounding pixels, the checkerboard is clearer, and the corners are easier to detect.

In addition, some poor IQ renderings as well as relatively perfect reference IQ renderings are listed in the appendix at the end of this article.

3.2. Calibration Steps

3.2.1 Model calibration

In the model calibration stage, each type of product only needs to be calibrated once. It is recommended to select one product from the same batch for model calibration according to the Three Sigma principle. In order to ensure the accuracy and robustness of the calibration results, a large number of images with checkerboards should be taken with this product.

3.2.1.1. Checkerboard calibration board

Prepare a checkerboard calibration board for calibration, the accuracy of the calibration board should be high, it is recommended to buy a professional calibration board. An example checkerboard calibration board is shown in Figure 5, and the specific specifications are as follows: the checkerboard calibration board has a total of 12*9 black and white squares, the inner corner points are 11*8, and the inner corner points are shown in the red circle in Figure 5. The side length (physical unit) of a single square is 95mm. Note: the scale of the calibration board in the **shot** image should not be too small. In general, the side length (pixel unit) of each square in the image should not be less than 10 pixels; If the calibration board occupies too few pixels in the picture, you can consider moving the calibration board close to the camera or replacing the larger size of the checkerboard calibration board (if necessary, reduce the number of squares in the checkerboard calibration board, the minimum number of inner corner points is 4*3). In addition, in the outermost circle of the customized professional checkerboard calibration board, the distance between the black and white squares and the edge of the calibration board should be larger (at least the side length of a single black and white square), that is, enough white should be left.

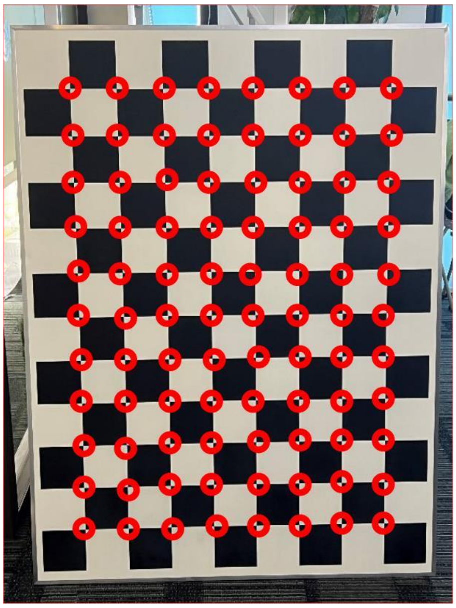

Illustration of a checkerboard calibration board5

3.2.1.2. cv server run

- Muffin, Maruko, Opera series
  - a) Modify the streaming json according to the board Settings and Table 1 configuration instructions, and the json is put into the commn directory
  - b) Put the common directory on the board side, such as /mnt/
  - c) Set the environment variable to specify the json path.export CV_SERVER_JSON_PATH=/mnt/
  - d) Type the following to run demo:./prog_cv_server
- Souffle series
  - a) Modify the streaming json according to the board Settings and Table 2 configuration instructions
  - b) Put./prog_cv_server_demo and json in the same directory on the board side
  - c) Type the following to run demo:./prog_cv_server

Note:

(1) cv_server and sdk must be paired

See muffin_json_sample.zip for json examples of Muffin, Maruko, and Opera chips

(3) Please refer to souffle_json_sample.zip for example of Souffle series chips

muffin_json_sample.zip souffle_json_sample.zip

Table 1 muffin cv server json configuration instructions1

Nodes	Configuration	Instructions	Notes
stPadVifInfo	snrNum	Number of sensors for the current video stream	Calibration captures are all 1
	snrResW	The width of the Sensor resolution	The resolution must be supported by the sensor

	snrResH	The Sensor resolution is high	The resolution must be supported by the sensor
	padId	SensorPad ID	
	GroupId	Vif Group ID	
	vifDevIdInGroup	The number dev in the Vif group	Generally not modified
	vifPortId	Vif's output port	Generally not modified
stIqCfg	IqFilePath	IQ file path	
stIspCfg	IspDevId	Isp device number	
	IspChnId	Isp channel number	
	IspOutPortId	Ispoutportid Output port number	
	Isp3DNRLevel	ISp3dnrlevel	
stSclCfg	SclDevId	Scl device number	
	SclChnId	Scl channel number	
	SclOutPortId	Scl output port number	
	SclOutPortW	Scl output width	Unit: pixel
	SclOutPortH	Scl output height	Unit: pixel
	SclHwId	Scl Hardware ID	
stVencCfg	vencDev	Venc device number	
	u32VencChn	Venc channel No	
	eType	Encoding type	2: H264 3: H265 4: JPEG

Table 2 souffle cv server json configuration description2

Nodes	Configuration	Instructions	Notes
stVifCfg	snrResIdx	Index in the Sensor resolution map table	
	padId	Sensor Device number	
	GroupId	Vif Group No	
	vifPortId	Vif's output port	Generally not modified
stIqCfg	IqFilePath	IQ file path	
stIspCfg	IspDevId	Isp device number	
	IspChnId	Isp channel number	
	IspOutPortId	Ispoutportid Output port number	

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	Isp3DNRLevel	ISp3dnrlevel	
stSclCfg	SclDevId	Scl device number	
	SclChnId	Scl channel number	
	SclOutPortId	Scl output port number	
	SclOutPortW	Scl output width	Unit: pixel
	SclOutPortH	Scl output height	Unit: pixel
stVencCfg	VencDev	Venc device number	
	VencChn	Venc channel No	
	eType	Encoding type	2: H264 3: H265 4: JPEG

3.2.1.3. Single camera calibration picture shooting

The picture taken of the scene is used to calibrate the camera internal reference. **When shooting, it is best not to hold up the checkerboard calibration board** (holding up the checkerboard will cause the internal corners to be blurred due to the slight shaking of the hand, which is not conducive to corner detection). Place the chessboard calibration board in front of the camera lens, and constantly change the direction, distance and position of the chessboard calibration board in the picture, so that it uniformly covers the whole picture and covers four distances. Close and single checkerboard calibration board covers the whole picture, close and multiple checkerboard calibration board covers the whole picture (checkerboard calibration board accounts for about 1/3 of the picture), medium distance and multiple checkerboard calibration board accounts for about 1/6 of the picture (checkerboard calibration board accounts for about 1/6 of the picture), and long distance and multiple checkerboard calibration board has different angles in the picture. The actual distance of the internal reference calibration is related to the field of view of a single lens. The smaller the lens field of view is, all the distances of the model calibration need to be increased accordingly. Otherwise, all distances need to be reduced accordingly. (Note: the calibration map of the closest distance must ensure that the corners are clear; If it is beyond the range of effective depth of field, it is not necessary to grab a close-up map with full coverage of a single calibration board)

- Naming rules for single camera calibration photos

  < fixed prefix name >< camera number >_< image number >.jpg/.yuv, for example: camera0_0.jpg, for the first image of the first camera; camera0_1.jpg, for the second image from the first camera; camera1_2.jpg, for the third image from the second camera.
- The detailed steps of single camera calibration image shooting are shown in Table 3

Table 3 Steps of single camera calibration picture shooting3

Steps	Goals	Calibration method	Chessboard placement method (picture in Figure 6 as an example)	Quantity (photos)
1	Close range, full coverage of a single calibration board	The checkerboard calibration board fills the whole frame in the image, accounting for about 1/1. Take two pictures from this position, the first one facing the lens and the other one based on the first one with the checkerboard center line as the axis and rotated 10 degrees inside.	1. Fill the whole image 2. Based on the first image, rotate about 10 degrees with the center line of the checkerboard as the axis	2
2	Close up, multiple calibration boards with full coverage	Cover the full screen (4 corners + middle of the picture, a total of 5 positions), each picture contains about 1/3 of the checkerboard calibration board, take two pictures at each position, the first one is as straight as possible to the lens, and the other one is tilted about 10° based on the first one (the four corners change in trapezoidal shape).	1. Place in the middle of the picture 2. Rotate about 10° with the center line of the checkerboard as the axis 3. Place in the bottom left corner of the screen 4. Rotate about 10° into the frame with the long side as the axis 5. Place in the bottom right corner of the frame 6. Rotate about 10° into the frame with the long side as the axis 7. Place in the top right corner of the frame 8. Rotate about 10° into the frame with the long side as the axis 9. Place in the top left corner of the frame 10. Rotate about 10° into the frame with the long side as the axis	10

Steps	Goals	Calibration method	Chessboard placement method (picture in Figure 6 as an example)	Quantity (photos)
3	Medium distance, multiple calibration plates full coverage	Double the distance between the calibration board and the camera, 4 corners + the middle of the picture, a total of 5 positions. In each picture, the checkboard calibration board takes up about 1/6, and two pictures are taken at each position. The first one is as far as possible in front of the lens, and the other one is tilted about 10° based on the first one	1. Place in the middle of the picture 2. Rotate about 10° with the center line of the checkerboard as the axis 3. Place in the bottom left corner of the screen 4. Rotate about 10° into the frame with the long side as the axis 5. Place in the bottom right corner of the frame 6. Rotate about 10° into the frame with the long side as the axis 7. Place in the top right corner of the frame 8. Rotate about 10° into the frame with the long side as the axis 9. Place in the top left corner of the frame 10. Rotate about 10° into the frame with the long side as the axis	10
	Long range, multiple angles	Continue to double the distance between the calibration board and the camera, and take two pictures at each position, with the other one tilted about 10 degrees from the first one. There is no position limit. However, the images taken at this distance should be taken from a different Angle.	1. Place in the middle of the frame 2. Rotate about 10 degrees with the center line of the checkerboard as the axis 3. It is placed in the bottom right corner of the screen 4. With the short side as the axis, rotate about 10 degrees inward into the frame 5. Place in the top right corner of the frame 6. Rotate about 10 degrees into the frame with the short side as the axis	6

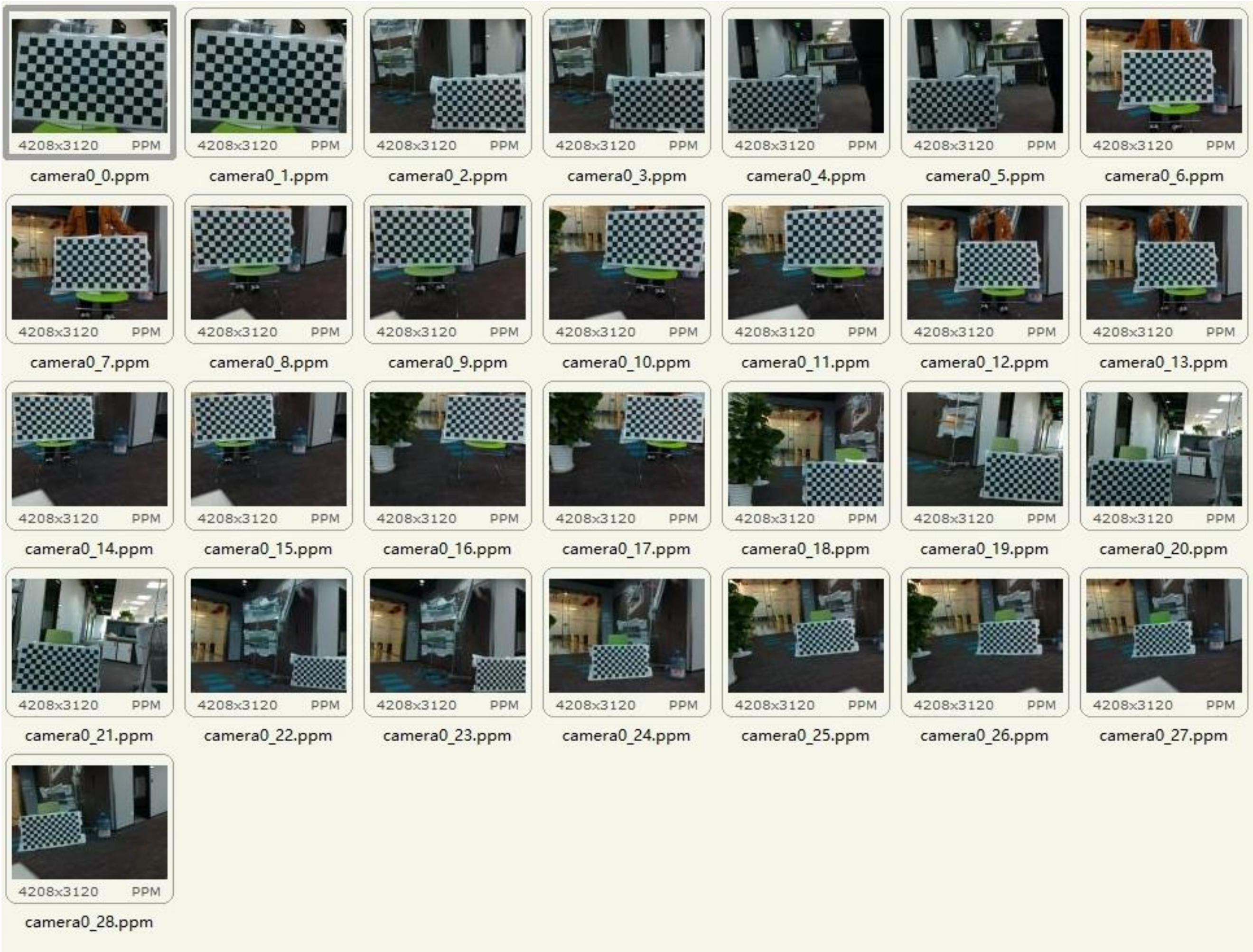


Image single camera calibration shooting example6

- Shooting tools

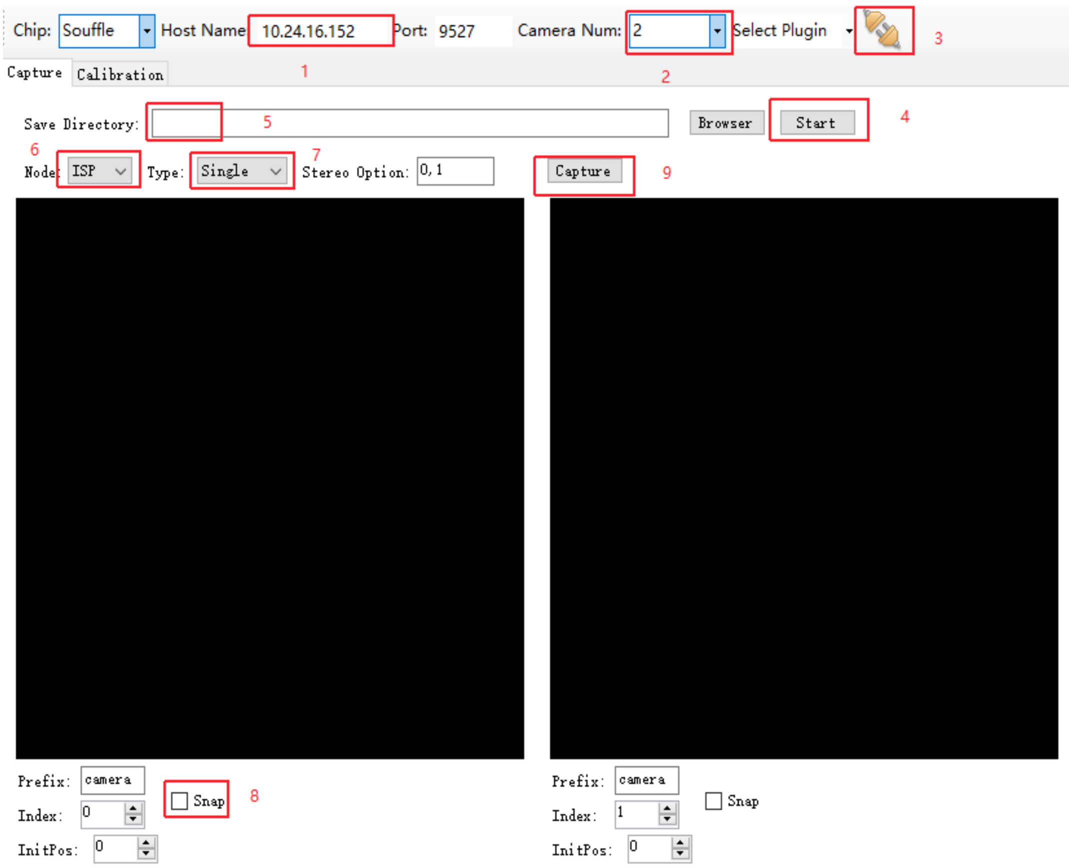
  1. Use third-party tools such as vlc, potplayer, etc. to watch the stitching effect in real time;
  2. Use SigmaStar's CVTool with prog_cv_server to capture images

     a) Run prog_cv_server on the board side;

     b) Connect prog_cv_server according to steps 1-3 of Figure 7;

     c) Click the Start button in step 4 of Figure 7 to start the preview;

     d) Grab the picture according to steps 5-9 of Figure



7.

Figure CVTool single camera capture steps7

See Table 4 for the CVTool interface configuration instructions:

Table CVTool capture configuration description4

Configuration	Instructions	Notes
Chip	Chip type	
Host Name	prog_cv_server The IP to run the board	Use with prog_cv_server
Port	prog_cv_server the port on which to listen	Default 9527
Camera Num	Number of cameras	Use with prog_cv_server and must match the number of previews created by prog_cv_server
Save Directory	Picture save path	Save to the English path, and the path can not contain space, decimal point and other special characters;
Node	Catch the graph of which node of the whole link.	VENC nodes only need to support RTSP preview, other capture nodes

	(There are five nodes VIF, ISP, LDC, SCL, VENC)	must be equipped with prog_cv_server
Type	The type of grab	Single: Single camera calibrated grab Stereo: Dual-camera calibration grab
Stereo Option	Stereo option 2 Number of both cameras when calibrating a capture	The number must match the configuration of the Index below
Prefix	The filename prefix of the crawl image,	Example: if the Prefix is camera, the image will be named cameraX_Y.jpg
Index	Camera number	This number is reflected in the capture file name, such as cameraX_Y.jpg, where X is the camera number
Snap	Grab switch	This camera will only support capture once Snap is checked
Capture	Grab button	
Start	Button to preview the grab	
InitPos	Initpos Image number start position	This number is reflected in the capture file name, such as cameraX_Y.jpg, where Y is the image number

3.2.1.4. *Double camera calibration picture shooting*

The calibration picture taken for the scene is used to calibrate the out-of-camera parameters. The checkerboard calibration board should never be held in the hand when shooting the double-camera calibration picture (note the difference with the single-camera calibration picture). Dual camera calibration picture shooting requires that the two cameras must be synchronized to capture the picture, that is, to ensure that the overlap area must be captured by the two cameras at the same time, so that each data point on the overlap area is one-to-one corresponding in the two cameras, so as to ensure the rationality and accuracy of dual camera calibration.

- Naming rules for dual-camera calibration pictures

  < fixed prefix name >< camera number >_< camera number 1>< camera number 2>_< image number >.jpg/.yuv, for example: camera0_01_0.jpg represents the first image of the first lens; camera0_01_1.jpg stands for the second image from the first shot; camera1_01_3.jpg represents the fourth image of the second shot.

- Double camera calibration picture shooting steps

Each set of cameras captures at least 18 pairs of pictures. When taking pictures, adjacent cameras should be determined first, and each group of adjacent cameras should capture pictures at the same time. When capturing, place the calibration board of the checkerboard in the overlap area, as close as possible to the lens, until the adjacent cameras have to see the complete checkerboard, then rotate the calibration board by a small amplitude of about 10° and take another group. Then gradually change the position of the checkerboard calibration board in the overlap area, and repeat the above steps to make the checkerboard calibration board cover the entire overlap area. Then change the distance to a further distance and take a few sets. Double camera calibration, checkerboard calibration board distance from the lens to have three: short distance, overlap area partial coverage (80%); Medium distance, overlapping area partial coverage (60%); Long distance and overlapping areas were partially covered (40%). The calibration distance of the external reference is related to the field Angle of view in the overlap area. The smaller the field Angle of the overlap area is, the distances calibrated by the external reference of the model calibration increase accordingly.

Table. Shooting method of dual-camera calibration pictures5

Steps	Goals	Calibration method	Chessboard placement method (For example, the picture in Figure 8 below, the number is the same as the file name, camera0, camera1 refers to different lenses, 0_01_0, 0_01_1... , 0_01_17 stands for different calibration images of the same lens; 0_01_n and 1_01_n (n is the index) represent the same pair of calibration maps for pairs of shots)	Quantity (pair)
1	Partial coverage of short distances, overlapping areas (80%)	The checkerboard calibration board should be as close to the lens as possible until the complete checkerboard can be seen by the adjacent cameras, starting from the top of the overlap area, after taking a pair of pictures	1. Place at the top of the overlap area; 2. Take the middle line of the checkerboard as the axis, rotate about 10°; 3. Move down to the middle; 4. Rotate about 10° with the center line of the checkerboard as the axis; 5. Continue to move down to the bottom of the overlap area; 6. The center line of the checkerboard was used as the	6

Steps	Goals	Calibration method	Chessboard placement method (For example, the picture in Figure 8 below, the number is the same as the file name, camera0, camera1 refers to different lenses, 0_01_0, 0_01_1... , 0_01_17 stands for different calibration images of the same lens; 0_01_n and 1_01_n (n is the index) represent the same pair of calibration maps for pairs of shots)	Quantity (pair)
		in the same position, the checkerboard should be rotated by about 10°. Then gradually move the checkerboard downward and repeat the above steps until you have moved to the bottom of the overlap area, at least three positions (top, middle, bottom).	axis and rotated about 10°	
2	Medium distance, overlapping area partial coverage (60%)	Approximately double the distance, starting from the bottom of the overlap area, after shooting a pair in the same position, the checkerboard rotates about 10°, then gradually move the checkerboard, repeat the above steps until moving	1. Place them in the middle of the overlap area; 2. Rotate about 10° with the center line of the checkerboard as the axis; 3. Move down to the lower end of the overlap area; 4. Rotate about 10° with the center line of the checkerboard as the axis; 5. Move up to the top of the overlap area; 6. Rotate about 10° with the center line of the checkerboard as the axis	6

Steps	Goals	Calibration method	Chessboard placement method (For example, the picture in Figure 8 below, the number is the same as the file name, camera0, camera1 refers to different lenses, 0_01_0, 0_01_1... , 0_01_17 stands for different calibration images of the same lens; 0_01_n and 1_01_n (n is the index) represent the same pair of calibration maps for pairs of shots)	Quantity (pair)
		to the top of the overlap area, there are at least three positions (top, middle, bottom), it is not necessary to strictly move from bottom to top during the shooting process.		
3	Long distance, overlapping area partial coverage (40%)	The calibration board is further away from the camera, starting from the middle and lower part of the overlap area. After taking a pair of pictures in the same position, the checkerboard calibration board is rotated about 10 degrees or so. Then move the calibration board up to the middle of the overlap area and the upper middle position.	1. Place in the middle position of the distal overlap area; 2. Take the middle line of the checkerboard as the axis, rotate about 10°; 3. Move down to the middle and bottom of the overlap area; 4. Take the middle line of the checkerboard as the axis, rotate about 10°; 5. Move up to the middle and top of the overlap area; 6. Rotate about 10° with the center line of the checkerboard as the axis;	6

Figure Binocular camera calibration picture shooting example8

Figure 6 and Figure 8 show the size, position and rotation state of the checkerboard calibration board in the calibration diagram. The diagram is only for guidance, and it is not necessary to be completely consistent in the actual calibration. The distance and Angle of the checkerboard calibration board can be adjusted according to the real camera used to achieve the above effect.

- Shooting Tools

  The steps are similar to single camera shooting, but configure the Type of CVTool tool interface to Stereo, and configure the Stereo Option.

- Precautions for photo shooting

  a) The calibration environment should be adequately illuminated and uniform as far as possible, otherwise the large environmental noise may affect the calibration (example figure is shown in the appendix at the end of the article);

  b) The checkerboard calibration board should not have obvious reflection/shadow and other places that lead to uneven light and dark, otherwise it will lead to corner detection. This can be avoided by adjusting the relative position of the camera, calibration board and light source (see appendix for an example image);

c) The checkerboard calibration board should not be too small in the picture. Generally speaking, the side length of each black and white square in the calibration board should not be less than 10 pixels. If the checkerboard calibration board occupies too few pixels in the picture, it may lead to the detection of internal corners or inaccurate detection of corners. Consider moving the calibration board closer to the camera or replacing the calibration board with a larger size (if necessary, reduce the number of checkerboard grids, the minimum number of checkerboard corners is 4*3);

d) There should be no more than one checkerboard calibration board or similar checkerboard format items in the screen, otherwise the corners will not be detected. If this occurs, please remove or block the item in advance, or manually paint out the part in the image editing tool;

e) The black and white squares of each checkerboard calibration board must be complete in the picture, otherwise the corners will not be detected; (See appendix for an example image)

f) The checkerboard in each captured image must be clear, and there must be no blur; If it is not clear enough, it will lead to the detection of corner pixel coordinates are not accurate enough, and then lead to the accuracy of calibration is not enough. Please first confirm whether the lens is clean, if not clean, wipe clean, and ensure that the wiping process will not move the lens focal length. In addition, check whether to adjust the focal length, to ensure that the focus is clear, and the depth of field is good, the whole calibration process is not allowed to focus/zoom the lens operation, calibration process if the focus, will lead to calibration parameters are invalid. Therefore, all calibration steps need to start again after focusing.

g) During the shooting, hands should not touch the black and white squares of the checkerboard in the calibration board, and the black and white squares of the checkerboard should not be occluded, otherwise the corners will not be detected (see the example picture at the end of the article);

h) All the positions of the checkboard in the shot should be combined to cover the whole picture (in the case of dual-camera calibration, it is to cover the entire overlap area);

i) After capturing the calibration image, set the optimal stitching distance and place the checkerboard at the overlap area of the paired cameras at this distance. The purpose is to check the calibration and stitching effect. If the checkerboard is not misaligned up and down, and there is no expansion or extrusion between the left and right, the calibration is accurate. If the checkerboard is misplaced and there is expansion and extrusion, the calibration is not accurate, and the possible reasons need to be analyzed.

3.2.1.5. Off-line splicing picture shooting

The off-line stitching pictures are used for verification and effect display after calibration is completed. They are only used for testing and have nothing to do with calibration. During shooting, the same set of calibrated cameras were used to take pictures corresponding to the number of stitching routes. If you want to better test the calibration and stitching effect, you can set the best stitching distance first, and place the checkerboard calibration board at the overlap of the distance (the linear distance between the camera and the lens) position. The purpose and function are shown in point i in the notes on image shooting at the top of this page.

- Naming rules for off-line stitching images

  < Fixed prefix name >< camera number >_stitch.jpg/.yuv, such as camera0_stitch.jpg

- Examples of pictures taken

Illustration of off-line stitching pictures9

3.2.1.6. Calibration

- Calibration Steps

a) Create an empty home directory, in this home directory create intr, extr empty directory, in extr directory create pair_i_i+1 empty directory, i represents the number of the camera;

b) Create the offline effect image directory in the main directory, such as: stitch to create stitch, distortion correction to create ldc, black light fusion to create nir, depth measurement to create dpu

c) Put the single camera calibration pictures into the intr directory;

d) Put the pictures taken by the double camera calibration into the extr directory under the pair_i_i+1 directory;

e) The images used to detect the stitching effect were placed in the stitch/ldc/nir/dpu directory according to the pattern.

f) Configure the corresponding parameters according to the steps in FIG. 8, and see Table 6 for the meaning of the parameters. If the parameters cannot meet the requirements, the Advance mode can be entered.

g) Click Run to start calibration;

h) After successful, the offline splicing effect will be displayed and the parameter file calib_out.json, parameter folder calib_para (used for production line calibration) and calib_parameter.txt (used for MI_LDC on the board) required for camera splicing will be generated in the Image directory. Where calib_out.json is automatically generated in the main directory after successful calibration, and calib_para is automatically generated in the main directory after successful calibration, which contains ori_model_intr.yml and ori_model_extr.yml files. Which store the internal and external parameters of the camera calibration respectively.

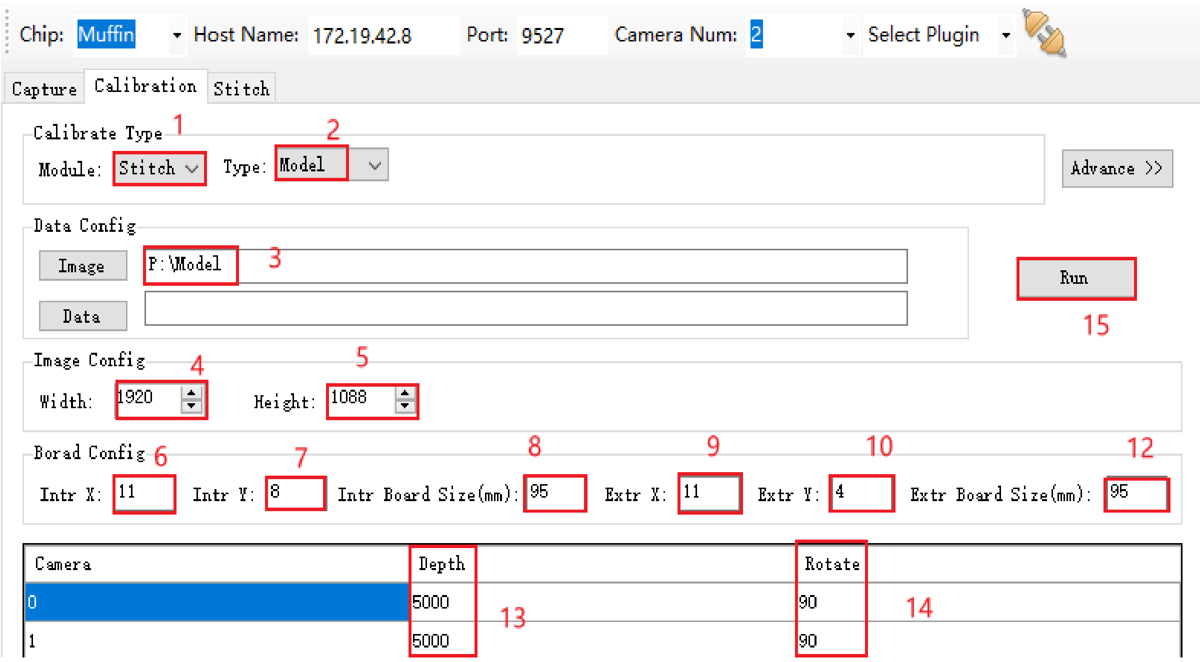


FIG. CVTool model calibration steps10

Table CVTool calibration configuration instructions6

Configuration	Instructions	Notes
Module	Calibrate the module used	To Stitch together;

		NIR: black light fusion; DPU: depth measurement; Single: Single camera calibration DPU_Stitch: Dynamic stitching LenSpec: Corrected using lens distortion parameters OneDimSingle: 1D single camera correction MultiLenSpec: Correction using zoom lens distortion parameters
Method	Single goal setting method	CheckBoard calibration LenSpec: Use lens distortion parameters to correct and fix focus MultiLenSpec: Use lens distortion parameters to correct, zoom
EIS	Electronic stabilization	single mode support for souffle chips only
Type	Calibration type	Model: model calibration; Plc: production line calibration; FineTune: fine tuning; Ptgui: panoramic stitching tool
FOV	Field of View	For Single->CheckBoard mode
Intensity	Distortion correction intensity	for Single mode, this parameter generates a new distortion table based on internsity
Lens Type	Lens type	Standard: normal lens; Wideangle: wide Angle lens; FishEye lens; WideangleEx: Wide Angle lens (extended)
Image	Calibrated home directory	There are intr, extr, stitch subdirectories under this directory
Data	The path of the calib_para folder, which is the path of the	This path needs to be set for production line calibration,

	internal and external parameter directories for the model calibration output	but not for model calibration
Image Format	Calibrate the type of image	RGB and YUV420 types are supported
Image Width	Calibrate the width component of the image resolution	
Image Height	Calibrate the high component of the image resolution	
Rotate	Rotate each lens by an Angle,	Only 0, 90, 270 are supported
Flip	Flip the picture left and right	
Mirror	Picture mirror flip	
Intr X	Single target set the number of inner corner points in the horizontal direction of the chessboard used for shooting	
Intr Y	Single target set the number of inner corners of the vertical direction of the chessboard used for shooting	
Intr Board Size	The physical length of a single black and white square in mm	
Extr X	Double target calibration the number of inner corner points in the horizontal direction of the checkerboard used for shooting	
Extr Y	Double target calibration the number of inner corner points in the vertical direction of the checkerboard used for shooting	
Extr Board Size	The physical side length of a single black and white square in mm	
Depth	Optimal stitching distance in mm	
Input Sequence Mode	Calibrate the sequence of the camera	Normal: positive order; Negative: reverse order
Separate Bin Mode	LDC Bin format	Overlap: The LDC output map includes the overlap area; No Overlap: LDC output map does

SigmaStar

		not include the overlap area
ISP Mode	ISP Input Mode	Horizontal: The images of input ISP are placed in the horizontal direction; Vertical: the images of the input ISP are placed in vertical direction
Check Re-Projection Error	Determine whether the reprojection error value is too large	Must be enabled
Plc Mode	Production line calibration mode	Intr&Extr: Calibration of internal and external parameters Extr: Calibrate only the external parameters
Projection	Projection mode	Linear: Linear projection Cylindrical: cylindrical projection Spherical: spherical projection Fisheye: fisheye projection Mercator: Mercator projection
Crop	Cutting methods	H: Horizontal way V: Vertical direction H&V: Horizontal and vertical directions
Blend->Percentage	Overlap area percentage	
Int Region Mode	Region of Interest mode	Manual: Manual mode Auto: Automatic mode
FovX	Horizontal FOV Angle	
FovY	Fovy 4 Vertically oriented field of view	
PrjCenterX	The projection center point is at the X-axis position of the output image	
PrjCenterY	The projection center point is at the output image Y-axis position	
Yaw	Yaw Angle, the Angle of rotation around the Y-axis	Range: -18000-18000
Pitch	Pitch Angle, the Angle of rotation around the X-axis	Range: -18000-18000

Roll	Roll Angle, rotation Angle around the z-axis	Range: -18000-18000
Crop	Whether to remove the black edges	Applies to Maruko chip Single mode Iford chip OneDimSingle mode
Out Width	Output image width after cropping	Works in Maruko chip Single mode Iford chip OneDimSingle mode
Out Height	Output image height after cropping	Works in Maruko chip Single mode Iford chip OneDimSingle mode
Correct Alpha	Horizontal tensile strength 1	Suitable for Maruko chip Single mode Iford chip OneDimSingle mode
Correct Beta	Horizontal tensile strength 2	For Maruko chip Single mode Iford chip OneDimSingle mode
Off X	Output image start X after cropping	Works in Maruko chip Single mode Iford chip OneDimSingle mode
Off Y	Output image start Y after cropping	Suitable for Maruko chip Single mode Iford chip OneDimSingle mode
Rotate	ISP rotation mode	Suitable for Maruko chip Single mode Iford chip OneDimSingle mode
Region Mode	Region mode for LDC mode	360_PANORAMA: 360-degree panorama mode for fish glasses in overhead and ground mounted modes 180_PANORAMA: 180-degree panorama mode for fisheye lens wall-mount mode NORMAL: Normal correction mode
Mount Mode	Mount mode for the Sensor	DESKTOP: Ground mount mode CEILING: Ceiling mode WALL: Wall mounted mode
Crop Mode	LDC zone cropping mode	NONE: Does not make any cropping FILING: Processing according to the principle of stretching, the valid area of the processed

		image will be cropped as much as possible STRETCH: According to the principle of equal proportion, the processed image is cropped and enlarged in accordance with the expected width and height ratio of the output image
In Width	Input image width	LenSpec and Single mode
In Height	Enter image height	LenSpec and Single mode
Out Width	Output image width	LenSpec and Single mode
Out Height	Output image height	LenSpec and Single mode
X Center Ofs	Horizontal offset of the image center point relative to the physical center point	LenSpec and Single modes
Y Center Ofs	The vertical offset of the image center point relative to the physical center point	LenSpec and Single modes
Radius	Image radius	LenSpec and Single mode
In Radius	The correction area corresponds to the inner radius of the original image	Only works in 360 panorama mode
Out Radius	The outer radius of the original image corresponding to the correction area	Only works in 360 panorama mode
Pan	Horizontal rotation Angle/starting Angle of the unfolded area	Indicates the starting Angle of the expanded area in 360 panorama mode
Tilt	Vertical rotation Angle/inner radius of the unrolled region	Indicates the inner radius of the unrolled area in 360 panorama mode
Zoom H	Image zoom/horizontal zoom scale/end Angle of the expanded area	In NORMAL mode, the zoom scale is indicated 180 Panorama mode indicates the horizontal zoom ratio 360 panorama mode indicates the end Angle of the expanded area
Zoom V	Zoom the outer radius of the scale/expand area vertically	Not valid in NORMAL mode 180 Panorama mode indicates vertical zoom ratio

		In 360 panorama mode, it indicates the outer radius of the expanded area
Rotate	Image rotation	Only works in NORMAL mode
Out Rotate	Output image rotation Angle	3
Focal Ratio	Used to describe the distance to the projection plane. The larger the value, the smaller the curvature, the smoother the picture. The smaller the value, the smaller the curvature, the more curved the picture	Valid in 180 and 360 panoramic mode
Distortion Intensity	Distortion intensity	
Pixel Size	pixel size of the Sensor	The unit is um*100 for zoom mode only
Fov H	Horizontal Fov	Zoom mode only
Fov V	Vertical Fov	Zoom mode only
Focal	Lens focal length	Single->CheckBoard mode only
Advance >>	Switch to advanced mode configuration	
<< Simple	Switch to the simple mode configuration	
Run	Start calibration	

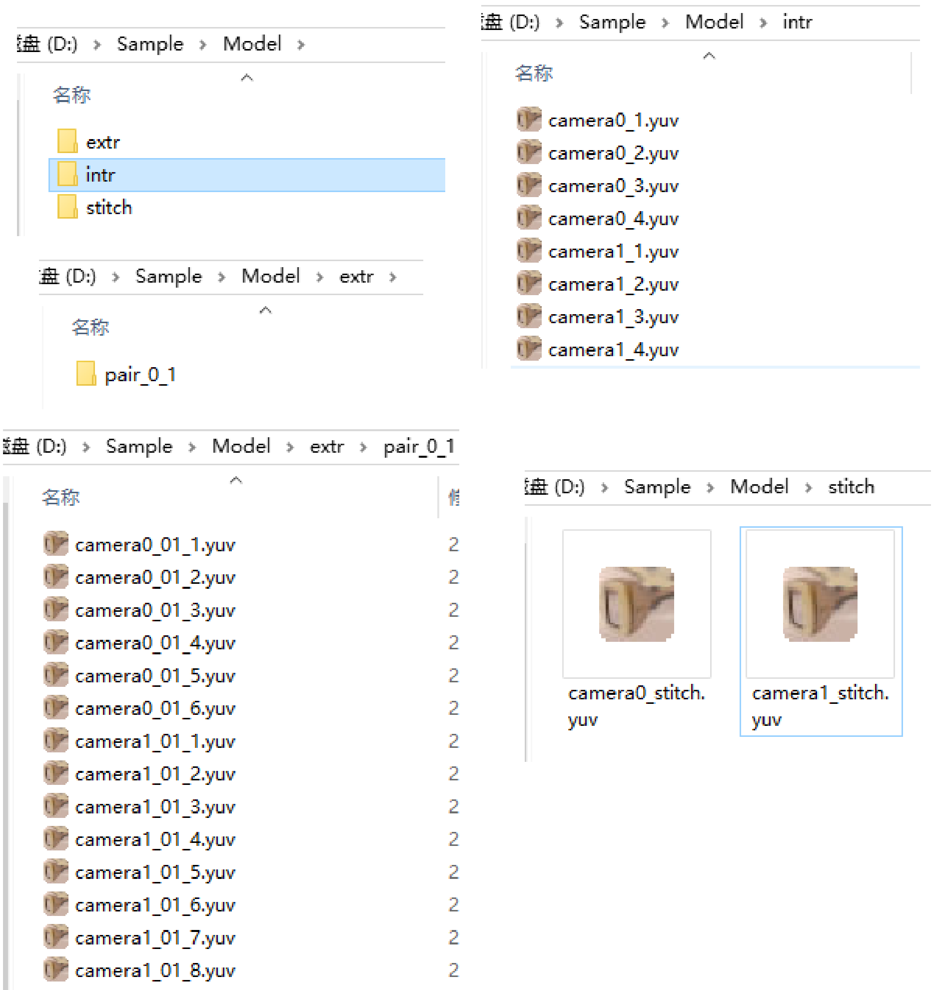


Figure catalog structure for binocular model calibration11

Rendering of the drawing before stitching12

Image model calibration stitching effect13

3.2.2 Production line calibration

After the laboratory calibrates the model of a certain individual (the same set of lenses and sensors) in the same batch, the final splicing effect will be affected by directly using the calibrated parameters of the individual model to splice other individuals due to production and assembly errors. Therefore, it is necessary to re-calibrate each individual in the same batch, adaptively adjust the internal and external parameters, and re-generate the bin file during production.

Our production line calibration scheme for static stitching requires only two images taken by a single camera and two pairs of images with overlapping areas. Dynamic stitching requires at least five sets.

- Checkerboard calibration board specifications

To improve the calibration accuracy, a custom checkerboard calibration board is recommended. Checkerboard calibration plate length of black and white squares (physical) is 60 mm, also need to meet each overlapping area inside the number of columns are greater than or equal to 3 columns. In the case of strong backlight and reflection, the checkerboard calibration board greatly affects the imaging effect of the checkerboard, thus affecting the calibration effect. Therefore, we must pay attention to the lighting of the production line environment, avoid backlight projection of the checkerboard calibration board, and do not appear strong lights around the checkerboard calibration board. The calibration board specifications we currently use are: 15 inner corner points (16 squares) wide, 8 inner corner points (9 squares) high, and 60mm side length. The total size of the calibration board is 640mm * 1100mm, as shown in the following picture:

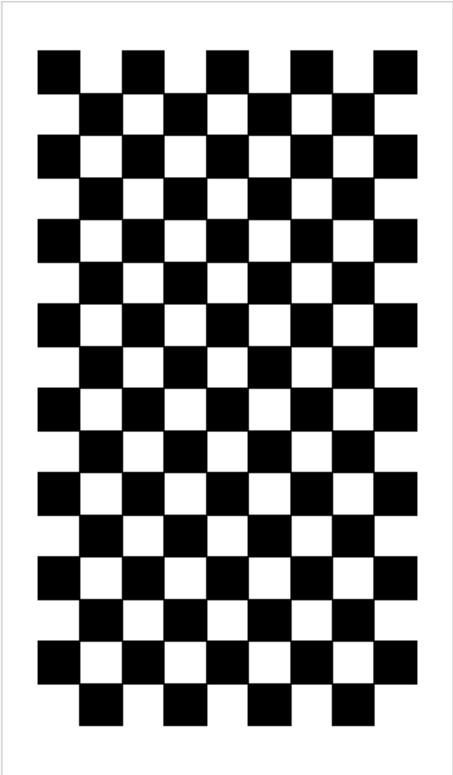
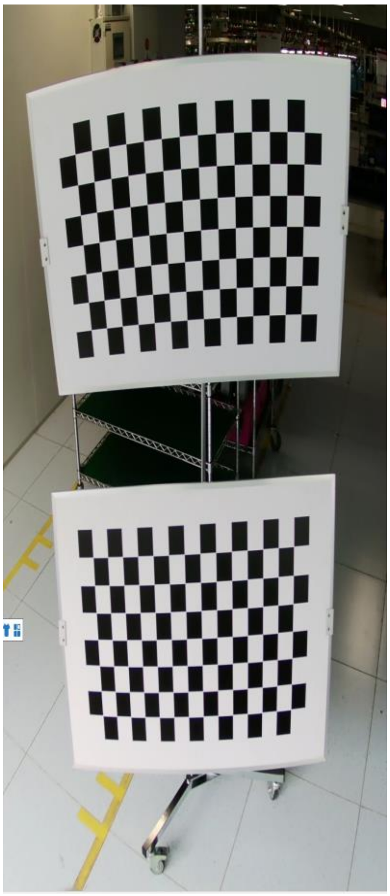


FIG. 14 Production line calibration checkerboard14

- Fix the checkerboard calibration board

The shelf for fixing the checkerboard calibration board is shown below, see the specific parameters marked in the figure,



and two calibration boards can be fixed on each shelf.

Diagram of fixed checkerboard calibration board15

- Placement

The schematic diagram of the shelf placement position of the fixed calibration board (four targets) is shown below. The middle four rectangles represent the four cameras, and the outermost red solid line represents the overlap area of two adjacent cameras, which is used for calibration of external parameters. The blue line represents the position of each camera, which is used to calibrate the internal reference. If the internal reference is not needed, it can be omitted. In four eyes, for example, a total of seven position the most outer ring (figure 7 solid line), each position 2 piece of checkerboard, a total of 14 pieces of checkerboard. For binocular (such as camera1 and camera2 in the image below), there are 4 positions (blue, red, blue, red), and 2 checkerboards above and below each position, for a total of 8 checkerboards. (Note: the checkerboard and the number of shelves are also related to FOV. If FOV is large, increase the number of shelves appropriately; If the FOV of camera1 in Figure 2 is large, add another shelf, add two new checkerboards, and put them on the leftmost side of the screen; The column may not be complete, but the row should be complete)

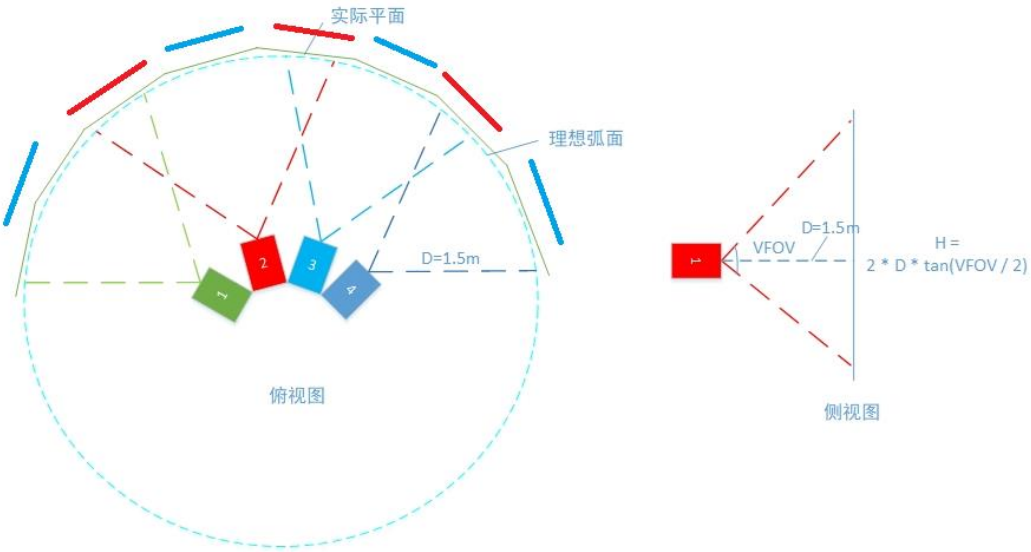


Figure production line calibration checkerboard layout16

- The distance between the shelf and the lens and the spacing between the two checkerboards in the shelf are related to the focal length of the camera, the FOV and the proportion of the overlapping area. It is necessary to ensure that the captured inner corners are clear and the number of inner corner points is sufficient (>=3). However, in order to ensure the coverage of the checkerboard in the vertical direction, the distance is also limited by the vertical height of the panel. The two checkerboards in the overlap area are 1.6m~2m away from the lens, and the vertical position is as symmetrical as possible. The other two positions (blue) are 1.6~2m away from the lens. In the preview picture captured, it should be reflected as close to the edge as possible, and the vertical direction should cover the whole picture as far as possible, and the horizontal direction should be as close to the edge as possible.

  - Production line calibration picture shooting requirements

    The images obtained by each camera should meet the following requirements:

    a) For single-camera and multi-camera calibration images, the column number of inner corner points should be greater than or equal to 3;

    b) The two checkerboard calibration boards in the vertical direction should be symmetrical up and down as far as possible, and it is best to cover about 80% of the entire overlap area.

    c) The more columns in the horizontal checkerboard detection, the better;

    d) Ensure that the inner corners of the checkerboard calibration board in the picture are complete in rows, and the columns are not complete;

e) The corners of the checkerboard are clear, and there is no obvious reflection/shadow that leads to uneven light and dark;

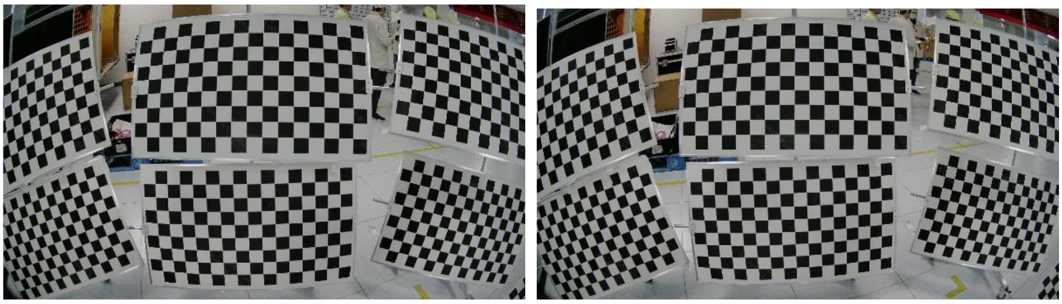
f) For external parameter calibration, for two adjacent checkerboard shelves, the spacing between the positive centers of the checkerboard must exceed 30% of the horizontal resolution.

g) The checkerboard should try to cover the four corners of the frame; The lens internal reference and distortion can be estimated more accurately;

h) The Angle points of the rows and columns of all the checkerboards in the picture should be consistent, and the configuration is the same as calib_in.json/CVTool;

i) The Angle between the normal vector of the checkerboard and the optical axis should be less than 45 degrees;

- Production line pictures taken
  a) Refer to the following picture to place the checkerboard
  b) After capturing a set of pictures, pan the lens back 50mm. Grab another set of pictures; (if doing dynamic stitching, at least 5 groups should be grabbed, that is, each group should move the module backward 50mm)
  c) Use vlc, potplayer or SigmaStar CVTool to capture images, and refer to single camera calibration for image naming.



Example of 17pictures taken by production line calibration

- Production line calibration steps
  a) Create an empty main directory for production line calibration inputs and outputs. In this home directory, create three empty directories: intr, extr, stitch. In extr, create an empty directory: pair_i_i+1, where i represents the number of the camera. And copy the output calib_para folder of model calibration to the main directory;
  b) Put the pictures taken by the single camera calibration of the production line calibration into the intr directory, one for each camera;

c) Copy the four pictures of the production line calibration and put them into the pair_i_i+1 directory of extr directory, where i represents the number of camera. Take the fourth mesh as an example, there are camera0_0.yuv,camera1_0.yuv,camera2_0.yuv,camera3_0.yuv in the intr directory. The extr directory has three subfolders pair_0_1, pair_1_2, pair_2_3, where the pair_0_1 directory stores camera0_0.yuv and camera1_0.yuv; The pair_1_2 directory stores camera1_0.yuv and camera2_0.yuv; Store camera2_0.yuv and camera3_0.yuv in the pair_2_3 directory.

d) Place the offline stitching effect test image in the stitch directory;

e) Configure the parameters according to the steps in Figure 18, and see Table 6 for the meaning of the parameters. If the parameters cannot meet the requirements, the advanced mode can be entered.

f) Click Run to start the production line calibration. After successful calibration, the offline splicing effect will be displayed and the splicing parameter files calib_out.json and calib_parameter.txt (the parameters used by MI_LDC at the board end) to be used for the chip will be generated.

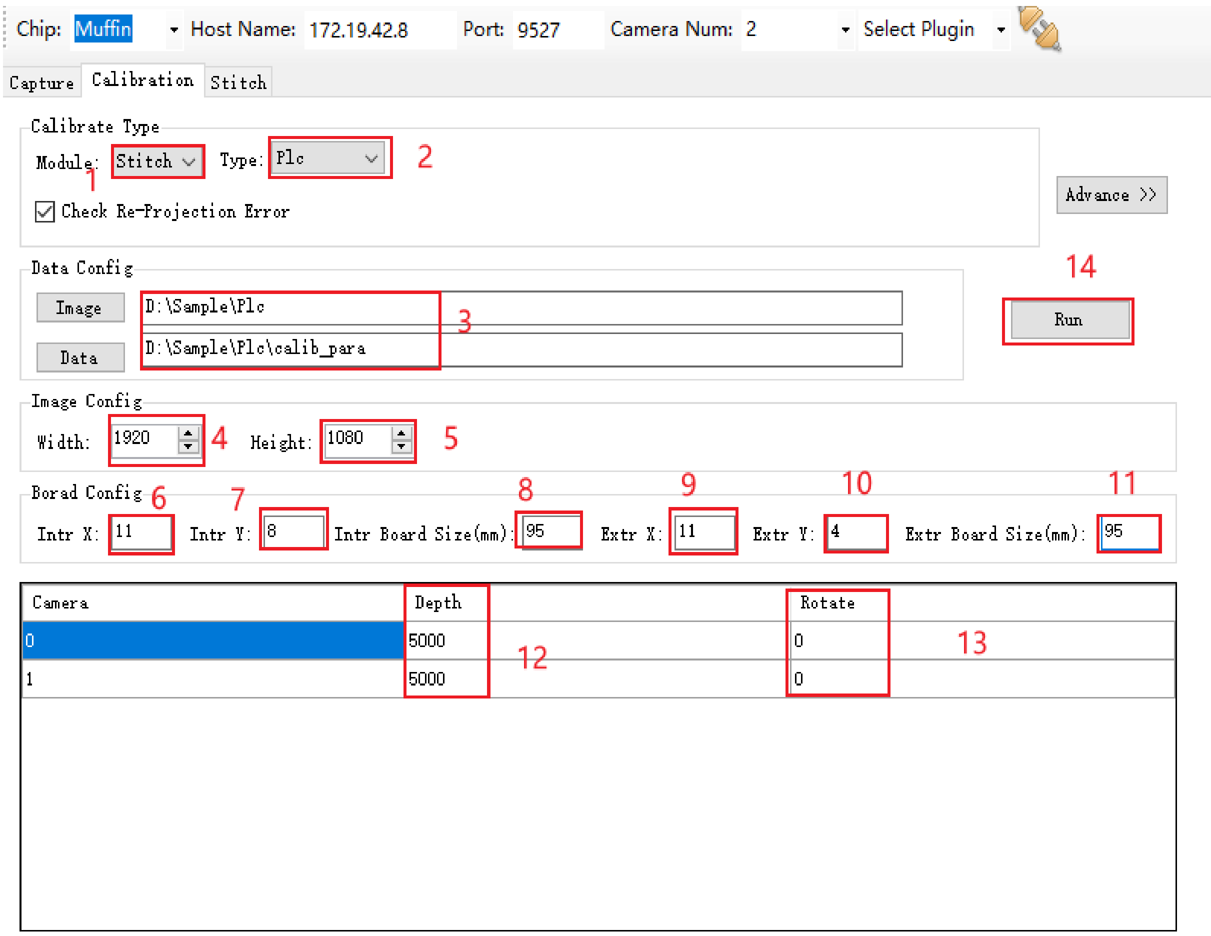


FIG. CVTool production line calibration steps18

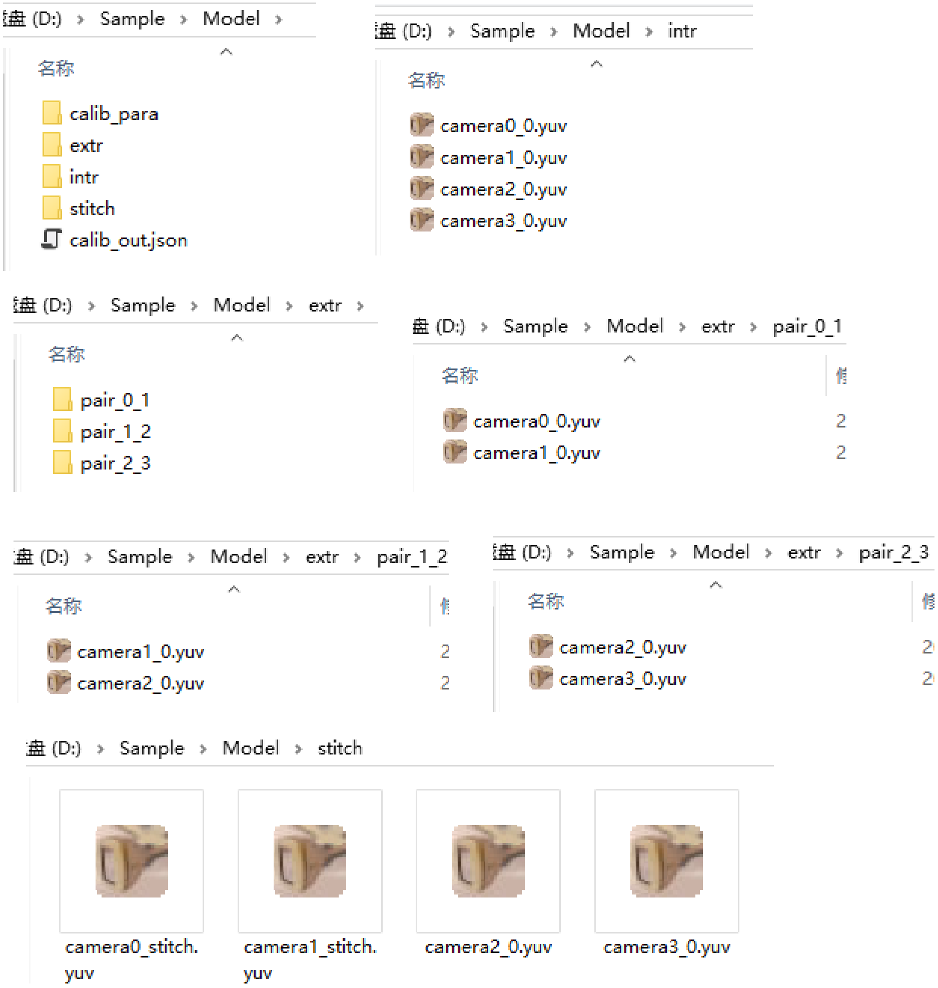


FIG. 4 Schematic diagram of catalog structure of production line calibration pictures19

FIG. lv mesh production line calibration effect diagram20

- Precautions for production line calibration:

  1. When shooting the checkerboard calibration board, pay attention to sufficient light and strong contrast of the picture;
  2. Ensure that the surface of the checkerboard calibration board is clean, and there is no reflective area on the surface of the checkerboard when shooting;
  3. Ensure that there is a certain spacing between the checkerboard calibration board to avoid two checkerboards being detected as a checkerboard;
  4. Software calibration failure, may be caused by the software background process, need to clean up the background process before re-entering the software;
  5. Pay attention to the influence of the fixture fixed to the checkerboard calibration board on the detection results, and the fixture should be fixed to the edge of the calibration board as far as possible, away from the black and white squares.

3.2.3 Auto Fine-Tune

After camera calibration, the structure of the camera may be offset due to environmental changes, transportation, etc., so the results of model or production line calibration may not be applicable anymore. If this phenomenon occurs, the Auto Fine-Tune function can be used to fine-tune, so as to correct the offset generated after calibration. Auto Fine-Tune takes one of the cameras as a reference and calculates the relative position relationship between the other cameras and the reference camera, so that the stitching misalignment is optimized. It should be noted that Auto Fine-Tune is only used as a support method after calibration, and it is optimized on the spot based on the calibration results, which has certain limitations.

- Shooting Requirements

  a) Simultaneous shooting with multiple cameras (only one picture per camera);
  b) Auto Fine - most cerebral sci-film optimization only smaller calibration dislocation, although larger mismatch can optimize, but the optimization effect is limited;
  c) Complex feature areas are required in the overlap area. Auto Fine - most cerebral sci-film depend on feature point detection, if the detection of feature points is not obvious or overlapping area more repeated characteristics may worsen the calibration results;
  d) The distribution of detail textures should be as uniform as possible to avoid excessive concentration in the same area.
  e) The image disparity is required to be uniform and the depth of field is more than 6 meters.

If the disparity is severely uneven, Auto Fine-Tune will cause partial misalignment.

- Photo shoot

  Use vlc, potplayer, or SigmaStar CVTool to simultaneously shoot (only one image per camera) according to the requirements, and refer to the single camera calibration for image naming.

Figure Auto Fine-Tune calibration picture example21

- Calibration Steps

  1. Create an empty home directory to hold the input and output of Auto Fine-Tune. in this home directory, create two empty directories: in and stitch;
  2. Put the calib_out.json output of production line calibration or model calibration in the in directory of the main directory;
  3. Put the offline stitching pictures in the stitch directory;
  4. The single images taken by each camera during Auto Fine-Tune calibration were put into the main directory, and the naming rules were the same as the naming rules of the single camera images taken by model calibration.
  5. Configure the parameters according to the steps in Figure 23, refer to Table 6 for the parameters, and enter advanced mode if the parameters cannot meet the requirements
  6. Click Run. After success, the offline splicing effect will be displayed and the splicing parameter file calib_out.json to be used by the chip will be generated.

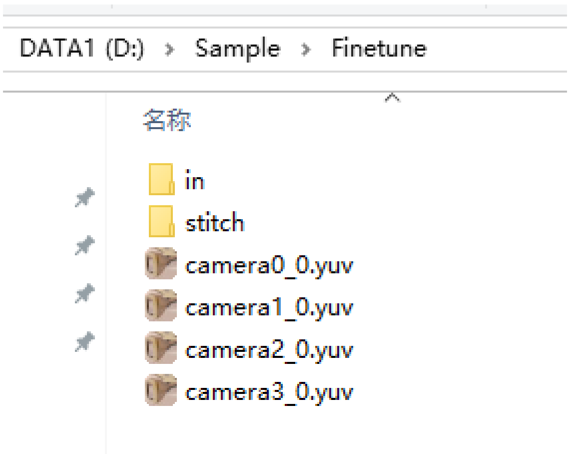


Figure Auto Fine-Tune directory structure22

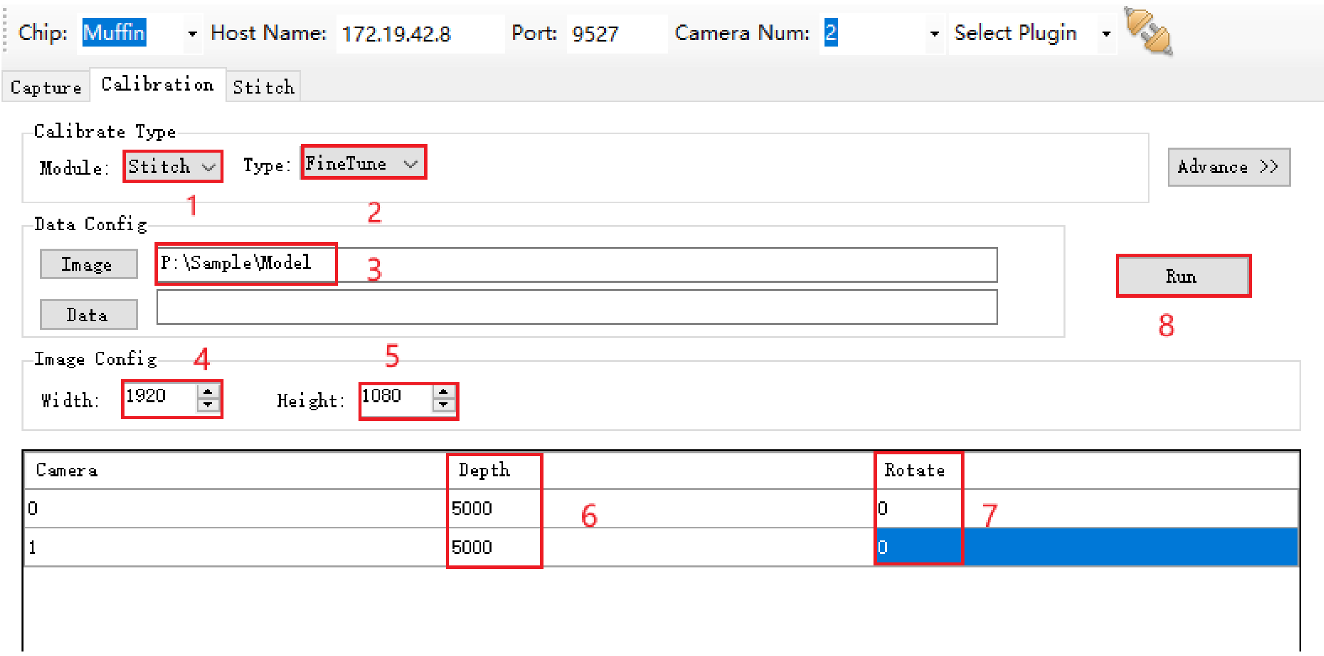


Figure the Auto Fine-Tune steps of CVTool23

- Auto Fine-Tune considerations
  a) The shooting pictures require rich features to avoid situations such as large areas of white walls in the image. If the number of feature points is too small (at least 4 pairs of matching points), an error will be reported and a re-shot will be required.
  b) Shooting distance more than 6 m, and the depth of the image as far as possible don't have too big change, namely taking pictures of scenery in a close distance;
  c) There is a separate object distance setting in the Auto Fine-Tune configuration file, that is, when Auto Fine-Tune calibration, the object distance taken by the camera is set at the place labeled 6 in Figure 23. Please be careful not to be confused with the object distance in the stitching diagram used for stitching.

3.2.4 Single camera calibration

3.2.4.1. Calibration based on a checkerboard calibration board

Single camera calibration is used to obtain the internal reference of the camera and correct the distortion of the camera according to the internal reference.

- Single camera calibration pictures are taken

  Single camera calibration is used to obtain the internal reference of the camera, and refer to 3.2.1.2 (Single camera calibration picture shooting) for picture shooting.

- Off-line effect picture shooting

  The effect diagram is used for the display of the effect after the distortion correction of the single camera. You can directly select from the single camera calibration picture or retake one and put it in the ldc directory. Name the picture camera_ldc.jpg/.yuv

  Note: If you take a single over-exposed image with a fisheye lens, name it camera_AE.jpg/.yuv and put it in the main directory

- Single camera calibration steps

  a) Create an empty home directory for the input and output of the single camera calibration. Create two empty intr and ldc directories in the main directory;

  b) Put the pictures taken by single camera calibration into the intr directory;

  c) Put the off-line effect pictures into the ldc directory, and see Figure 24 for the directory structure of single camera calibration;

  d) Configure the parameters according to the steps in Figure 25, and see Table 6 for the meaning of the parameters. If the parameters cannot meet the requirements, the advanced mode can be entered;

  e) Click Run to start model calibration. After successful calibration, the offline splicing effect will be displayed and the splicing parameter file to be used by the chip will be generated.

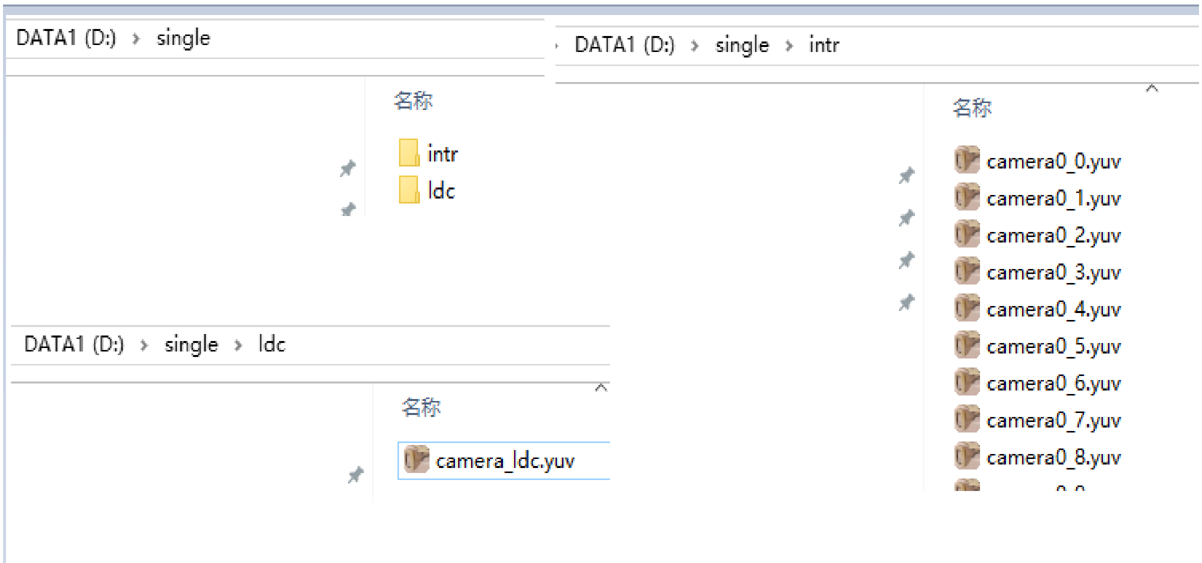


FIG. Single camera calibration directory structure24

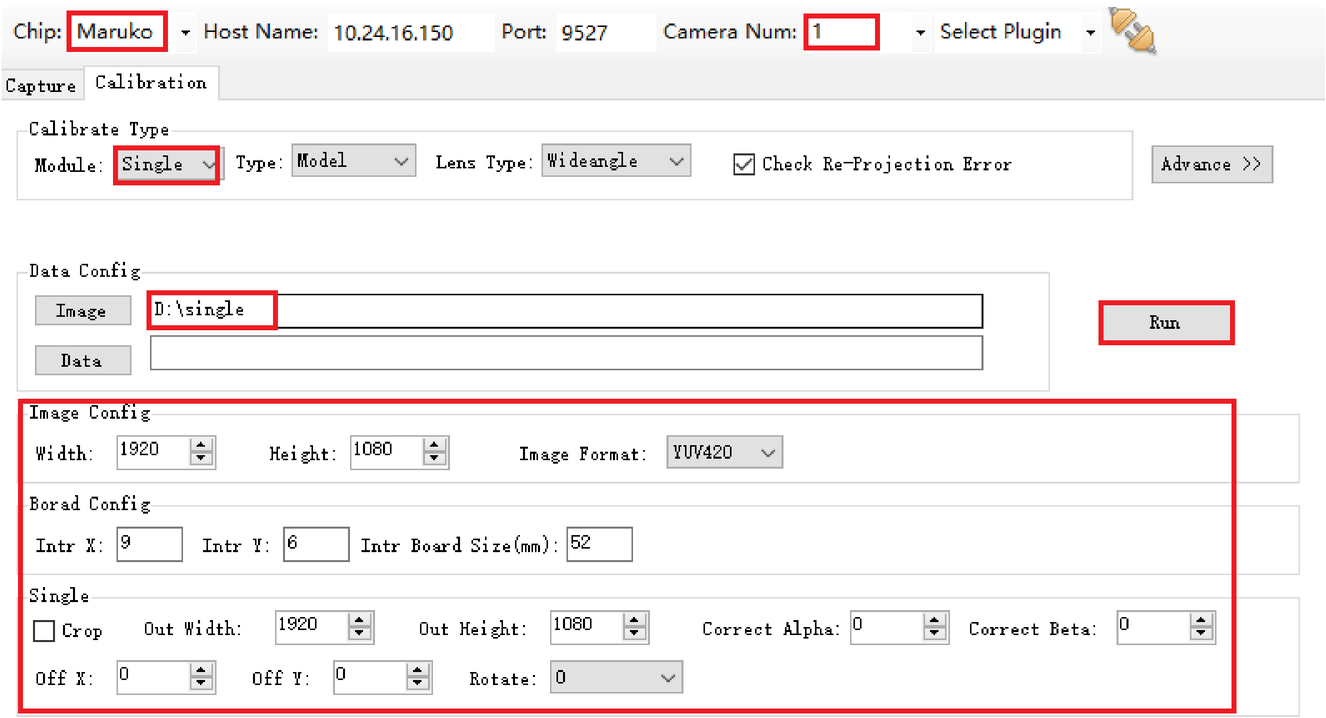


FIG. Maruko series chip single camera calibration steps25

Note: The steps of OneDimSingle mode of Iford series Chip are the same as above, only need to change the chip to Iford and the Module to OneDimSingle on the tool

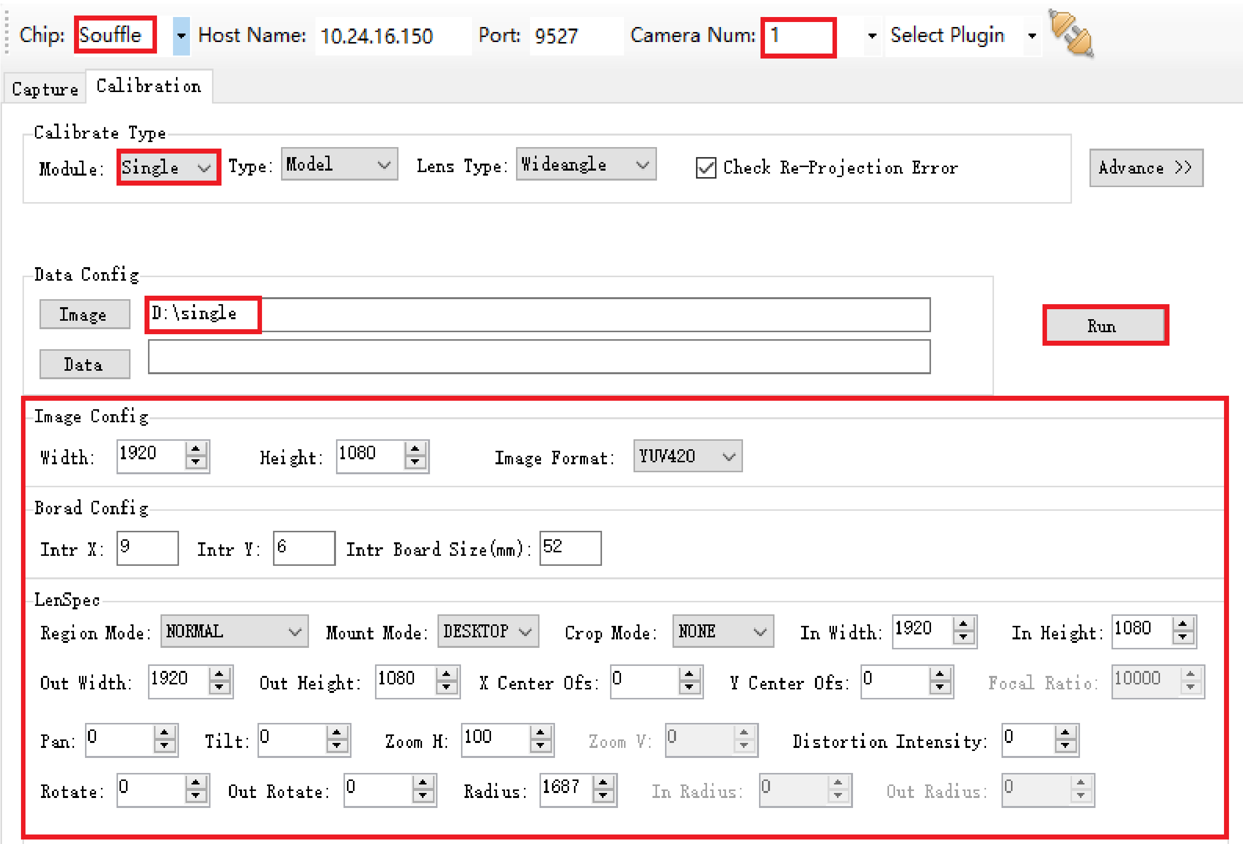


FIG. 26 Single camera calibration steps for Souffle series chips26

Note:

(1) The Maruko series generates calib_out.json in the home directory

(2) The Opera series generates mapX.bin and mapY.bin in the home directory

(3) The Souffle series generates CalibPoly_new.bin under poly/out

(4) To get the normal off-line correction effect, the Souffle series chip also needs to backfill xc, yc, radius in the poly/out/LenSpec.dat generated by calibration to X Center Ofs, Y Center Ofs, Radius, respectively. The calculation method is as follows:

X Center Ofs = xc – In_Width / 2

Y Center Ofs = yc – In_Height / 2

Radius = radius

(5) Iford series Chip Single mode operation steps are the same as Souffle, just change chip to Iford

3.2.4.2. LenSpec Way

LenSpec mode supports x, y map or CalibPoly_new.bin for distortion correction based on lens parameters.

- LenSpec.data preparation

  a) Parameter rows are separated by spaces, and distortion table rows are separated by tabs \t

Angle	real_height	Ref_height	xc	yc	radius	FOV	image_width	image_height
0.000	0.000	0.000	1280	720	1468	114	2560	1440
0.681	0.040	0.040						
1.362	0.079	0.079						
2.043	0.119	0.119						
2.724	0.159	0.159						

b) The table length should be aligned with the FOV, and the maximum angle value in the first column should not exceed 1/2 FOV

As shown in the figure below, the FOV is configured as 114, and the maximum angle value cannot exceed 57

Angle	real_height	Ref_height	xc	yc	radius	FOV	image_width	image_height
0.000	0.000	0.000	1280	720	1468	114	2560	1440
48.346	2.586	3.751						
49.027	2.615	3.842						
49.708	2.645	3.935						
50.389	2.673	4.032						
51.070	2.702	4.131						
51.751	2.730	4.233						
52.432	2.758	4.338						
53.113	2.786	4.446						
53.794	2.813	4.558						
54.475	2.840	4.673						
55.156	2.867	4.793						
55.837	2.893	4.916						
56.517	2.919	5.044						

- Off-line effect picture shooting

  The effect diagram is used for the display of the effect after distortion correction. It can be used to shoot a normal lens and put it in the ldc directory. The picture is named as follows: camera_ldc.jpg/.yuv

- Mode of Operation

  a) Create an empty home directory and in that home directory create the ldc directory

  b) Put LenSpec.data under your home directory

  c) Put the offline effect pictures into the ldc directory

  d) Follow the steps in Figure 28 or 29 to configure the parameters. See Table 6 for details on the meaning of the parameters

  e) Click Run, after success, generate the corresponding file in the directory

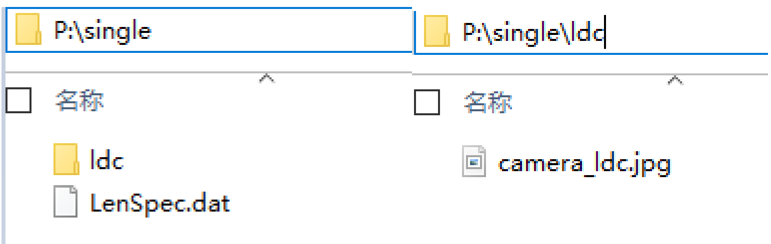


Figure 27 LenSpec directory structure27

Note:

(1) Opera series chips generate mapX.bin, mapY.bin, and map_info.txt

(2) Non-opera series chips generate CalibPoly_new.bin in the poly/out/ directory

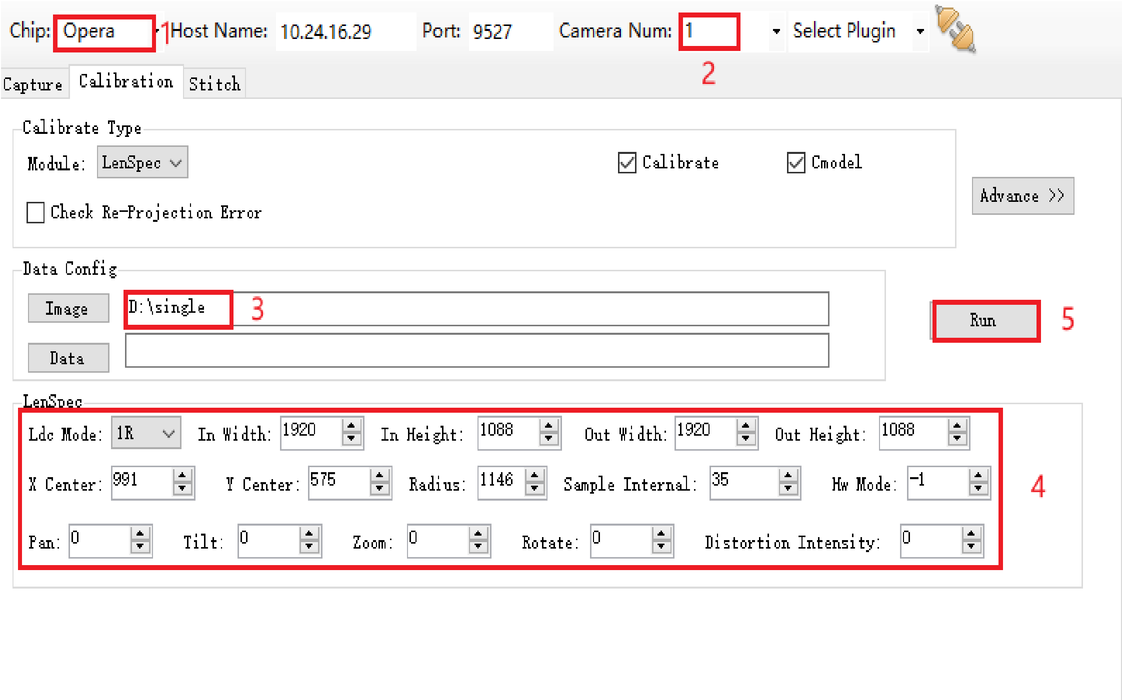


Figure LenSpec calibration method for Opera series chips28

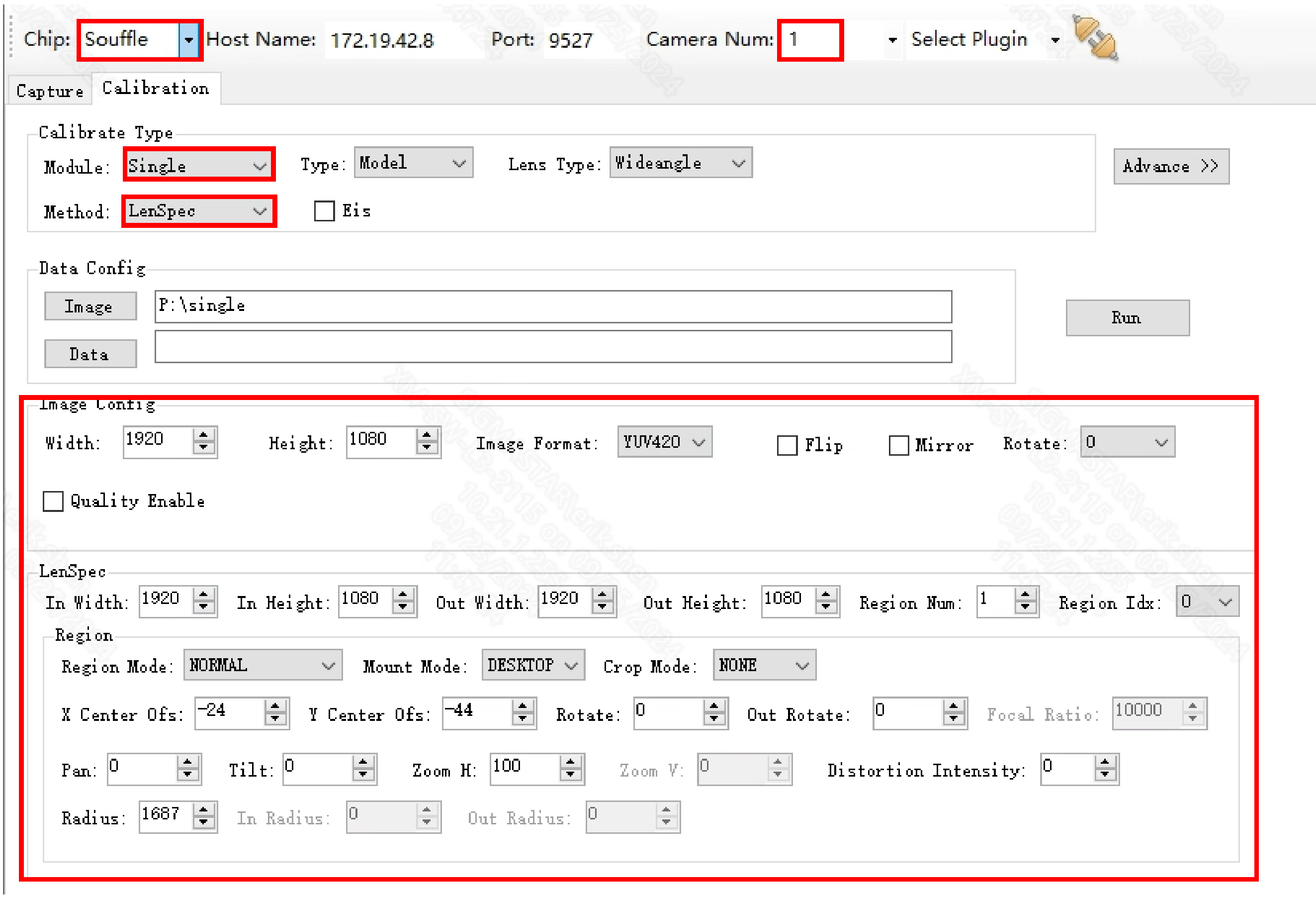
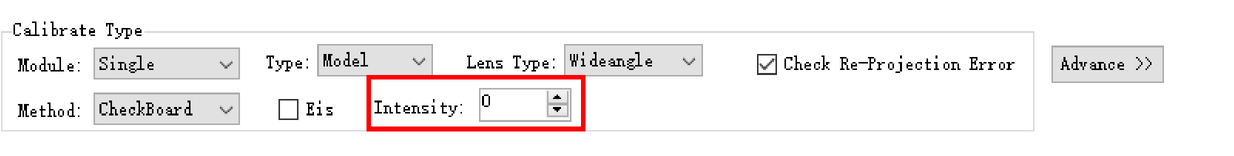


FIG. 29 LenSpec calibration method for Souffle series chips29

Note:

- Iford series chips are calibrated in the same way as Souffle, just change Chip to Iford.
- Souffle series chips support multi-region emulation, which can be switched by configuring Region Num and Region Idx.
- If you want to generate new distortion tables based on Intensity, set the following parameters. The path of the new distortion table after non-0 is configured is poly/out/LenSpec.dat



3.2.4.3. MultiLenSpec mode

MultiLenSpec mode supports the generation of CalibPoly_new.bin for distortion correction according to different focal length lens parameters.

- Name the LenSpec.data of different focal lengths as lenspec.dat, LenSpec_2.dat, LenSpec_x.dat in order
- Mode of operation
  a) Create an empty home directory such as MultiLenSpec and create the ldc directory inside that directory
  b) Put all LenSpec.dat in the directory
  c) Put the offline effect pictures in the ldc directory
  d) Tools configure the MultiLenSpec directory, as shown in Figure 31
  e) Click Run and generate CalibPoly_new.bin under poly/out in your directory

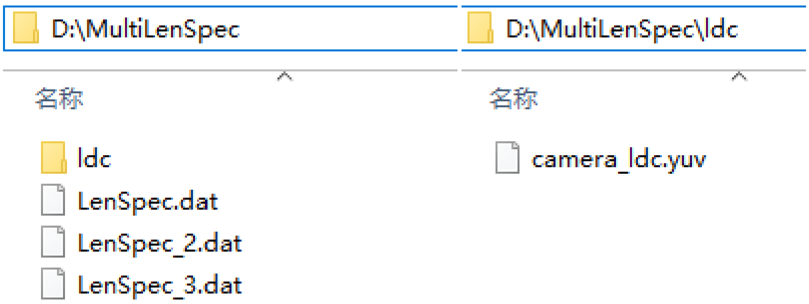


Figure 30 MultiLenSpec directory structure30

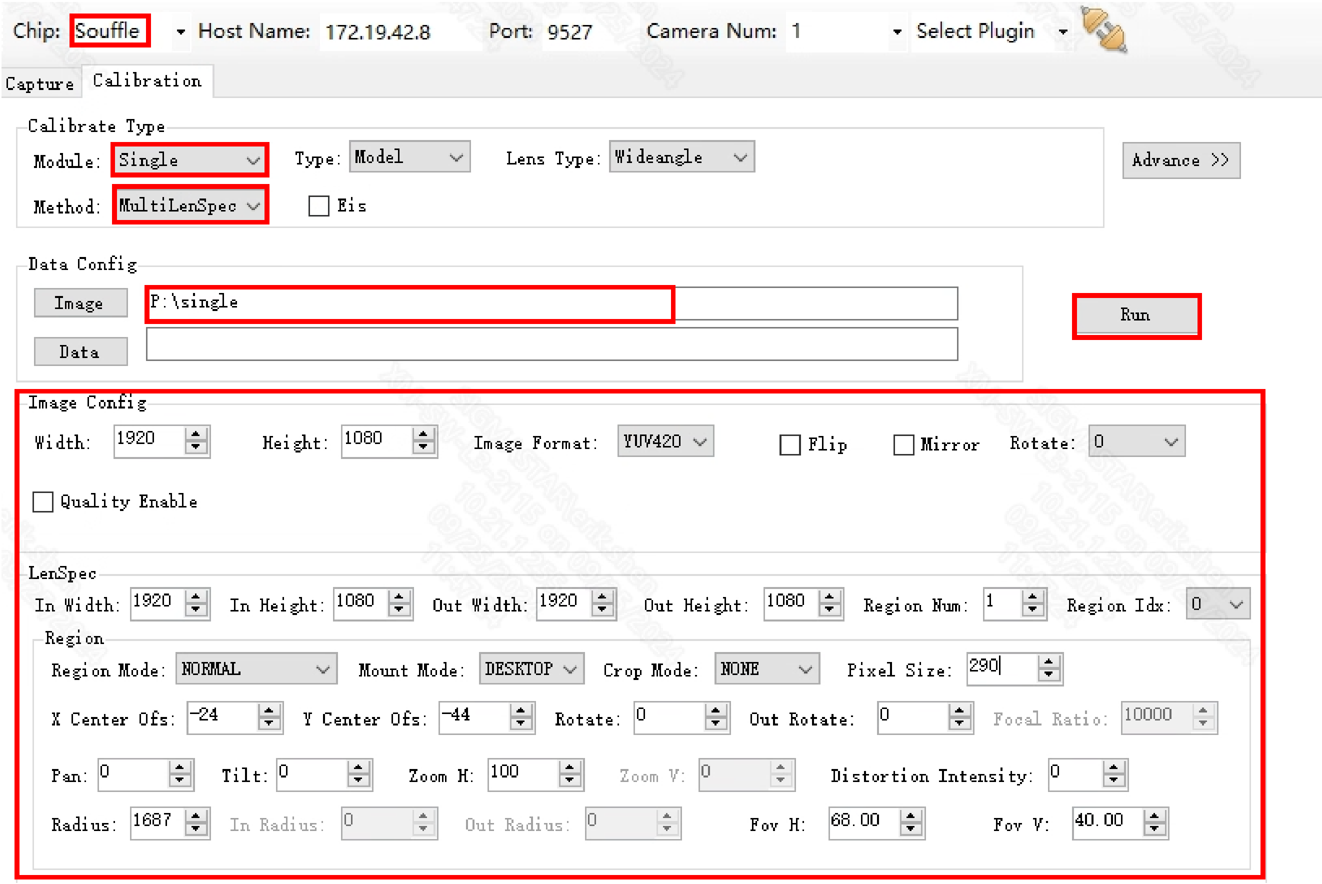


Figure 31 MultiLenSpec calibration method31

Note: Currently, only Souffle series chips of MultiLenSpec support offline effect display

3.2.5 NIR calibration

NIR calibration calculates internal and external parameters by taking checkerboard pictures, and fuses the same image acquired by VIS and NIR into a new image.

- Dual-camera calibration pictures were taken

  Double camera calibration is used to obtain the camera internal reference, and the picture is taken according to 3.2.1.4 (double camera calibration picture shooting). However, it should be noted that the overlap area of paired cameras calibrated by NIR basically accounts for all of a single picture, that is, if the entire overlap area is covered, the entire picture is covered. Therefore, the distribution of the checkerboard in the calibration image should be the whole picture (overlap area);

- Single camera calibration picture was taken

  The single camera calibration picture is used to obtain the internal reference of the camera. Due to NIR calibration, the overlap area basically accounts for the whole image. Therefore, double camera calibration is done first, and then the collected calibration map is directly copied to the intr directory as a single camera calibration map.

- Off-line effect picture shooting

  The effect picture is used for the corrected effect display. It can be directly selected from the double camera calibration picture or two shots can be taken at the same time and put in the nir directory. The picture is named cameraX_Y_nir.yuv, X is the lens number, Y is the number of photos, at least 2 photos of NIR offline effects are required.

- Calibration Steps

  a) Put the pictures taken by the dual camera calibration in the extr directory under the pair_0_1 directory

  b) Put the single camera calibration pictures in the intr directory

  c) Place nir cmodel pictures in the nir directory with a minimum of two per sensor

  d) The relevant parameters are configured according to the corresponding steps in Figure 33, and the meaning of the parameters is shown in Table 6

  e) Click Run to start the calibration

  f) After success, generate the camera's NIR parameters calib_out.json in the Image configuration directory, and generate NIR_out_stream.jpg in the nir_cmodel directory (cmodel effect diagram).

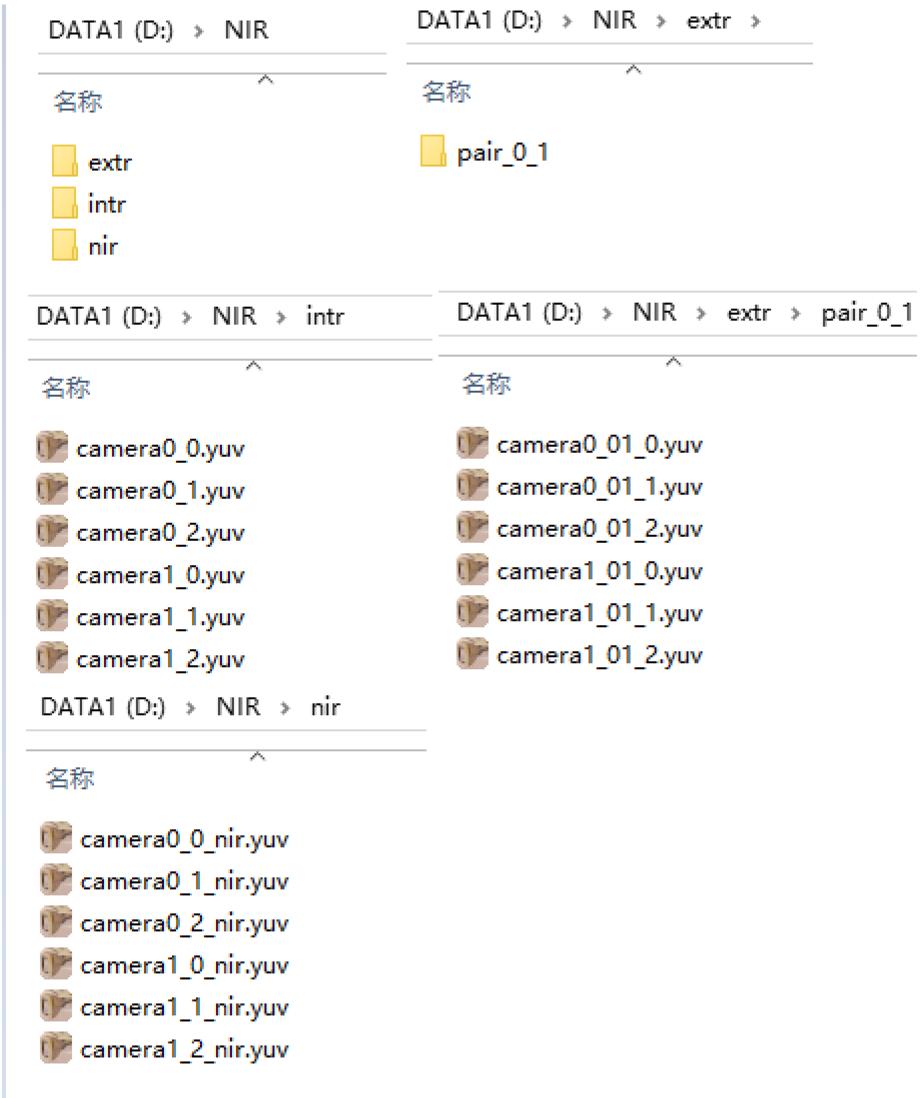


FIG. 32 NIR calibration image directory structure32

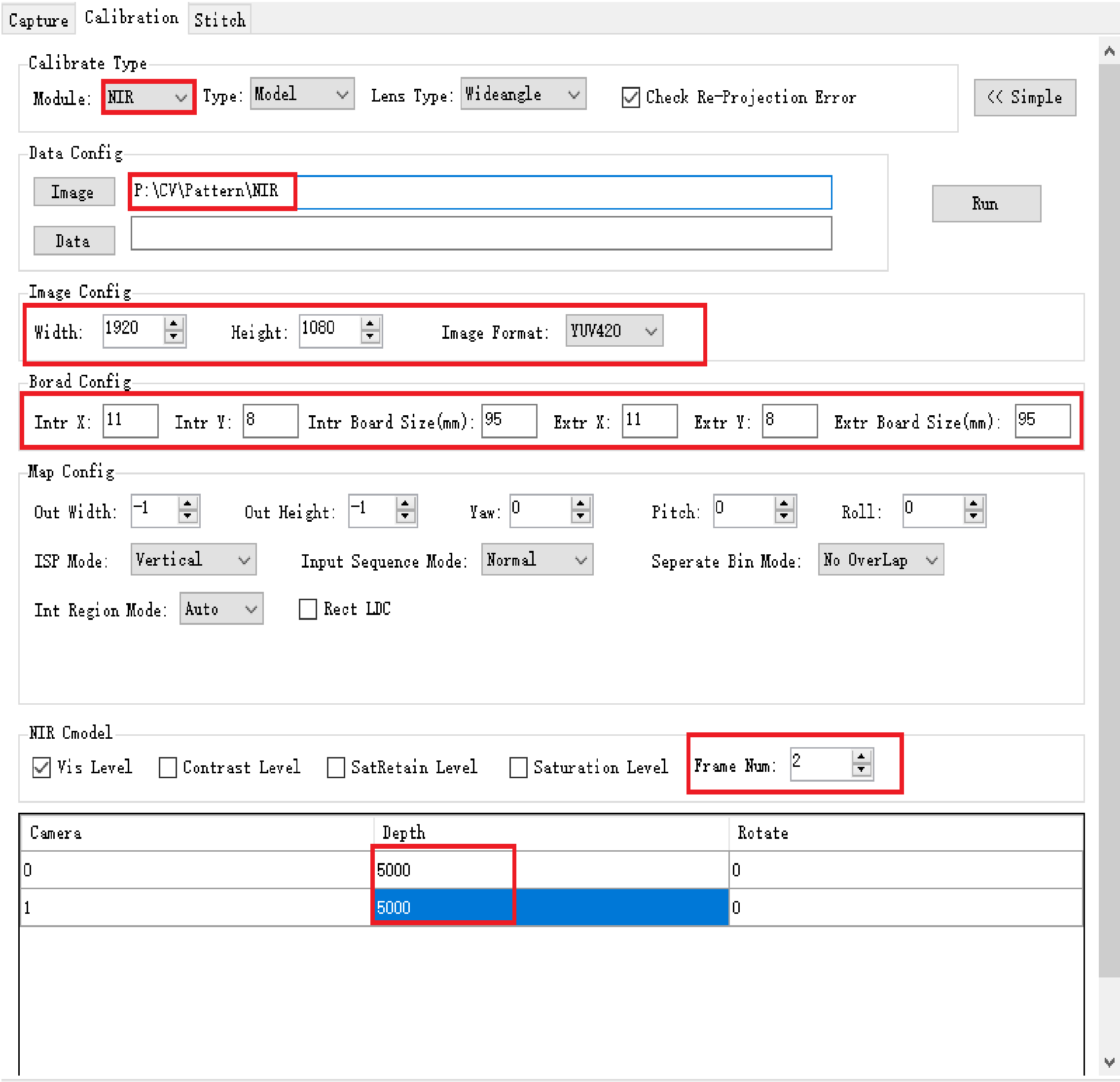


FIG. NIR calibration steps33

3.2.6 calib_out.json parameter description

After calibration with CVTool, the calib_out.json file will be output in the home directory. calib_out.json is divided into two parts, CalibInfo and mapgen. Tables 7 and 8 below illustrate the parameters of CalibInfo and mapgen, respectively.

- CalibInfo **(all parameters and parameter values in this part cannot be modified)**

Table CalibInfo parameter description7

Parameter	Definition	Notes
module	Calibrate the scene to be used	To Stitch together; NIR: black light fusion; DPU: depth measurement; Single: Single camera calibration

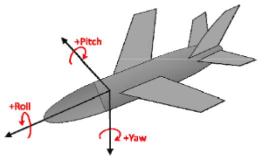
inputType	Calibration type	0: model calibration; 1: production line calibration; 3: Finetune
imgWidth	Calibrate the wide component of the picture resolution	
imgHeight	Calibrate the high component of the picture resolution	
imgDepthList	Set the optimal stitching distance in mm	There are several numbers in a few eyes Eg. If the optimal stitching distance is 2 meters, then ● Binocular is "2000,2000,"; ● Sanmu is "2000,2000,2000,";
rotateAngleList	Rotate each lens by an Angle. Currently 0, 90, 270 are supported This parameter can be set by looking at the calibration map. How many degrees is needed to rotate the calibration map clockwise to the positive screen, which is the Angle of rotation	There are several numbers in a few eyes Eg. If the calibration image captured by two lenses needs to be rotated 90 degrees clockwise to correct, set it to "90,90,";
camNum	Number of single shots	Eg. Binocular is 2
camIDList	Camidlist of lenses' ids	Binocular is "0,1,"; Trinocula is "0,1,2,";
camPairNum	Campairnum 2 Log multiple shots	Binocular means 1 pair, trinoces means 2 pairs
camPairIDlist_0	The first list of paired shot ids	Eg. For trinocles, for example, "0,1,"
camPairIDlist_1	A second list of paired lens ids	Eg. For trinocles, for example, "1,2,"
camOffxList	Both are 0 by default and do not need to be changed	
camOffyList		
imgEdgeTopList		
imgEdgeBottomList		

SigmaStar

imgEdgeLeftList		
imgEdgeRightList		
mapProjType	Projection type	0: linear projection; 1: cylindrical projection; 2: spherical projection; 3: fisheye projection; 4: Mercator projection;
mapCropType	Crop type	0: no cropping; 1: horizontal way; 2: vertical direction; 3: horizontal and vertical;
optimalFocal	Optimal focal length	Calibrate the estimated value

- mapgen

Table mapgen parameter instructions8

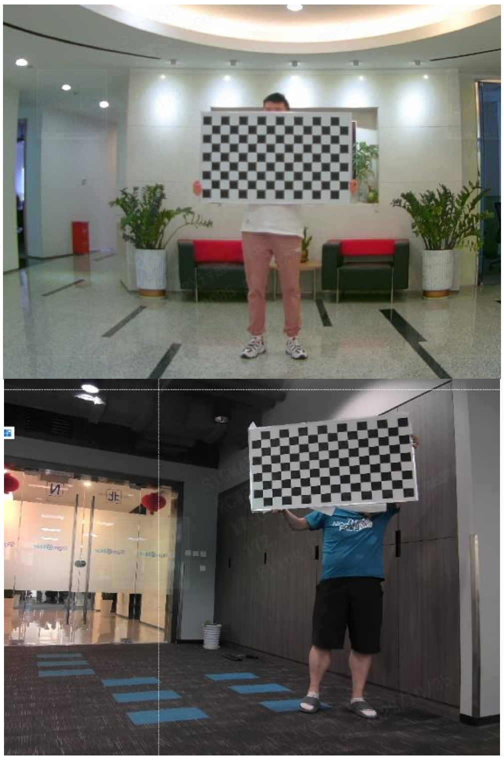
Parameter	Definition	Notes
scaleWidth	Output image width	● If only one of the two parameters is -1, the value of the other parameter is automatically adjusted; ● If both values are set to -1, the values are automatically calculated by the algorithm.
scaleHeight	Output image height	
in_seq_mode	Input image order in memory	0: Negative; 1: Normal;
src_mode	The way the input image is arranged in memory	0: vertical; 1: horizontal direction;
yaw	Yaw 4 Rotate along the X-axis in 3D coordinates	The adjustment of the attitude Angle in three directions is used to adjust the position of the image in the projection plane 
pitch	Rotate along the Y-axis in 3D coordinates	
roll	Rotate along the z-axis in 3D coordinates	
autoCalcMode	Whether the valid area image is automatically calculated	1: Automatically; 0: manual;
FOVX	Fovx horizontal FOV of	autoCalcMode is 0 and the output

	the output image	image size, i.e. scaleWidth and scaleHeight, must be set at the same time.
FOVY	Fovy The FOV of the output image in the vertical direction	
prjCenterX	Project center point X	
prjCentery	Project center point Y	
camRotation	The algorithm is used internally and does not need to be modified	
outputPath		
gridSize		
seprate_bin_mode		
preCropType		
ovp_capacity		
ovp_max_width		
updateBinMode		
hw_mode		
VerticalNum		
out_seq_mode		

Appendices

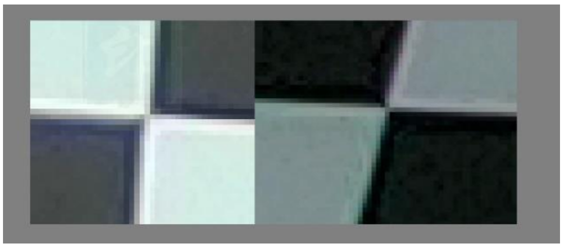
- This section shows several sets of bad IQ renderings, as well as one perfect IQ renderings;

1. Blurred checkerboard picture quality; (top: blurry quality, bottom: clear quality)

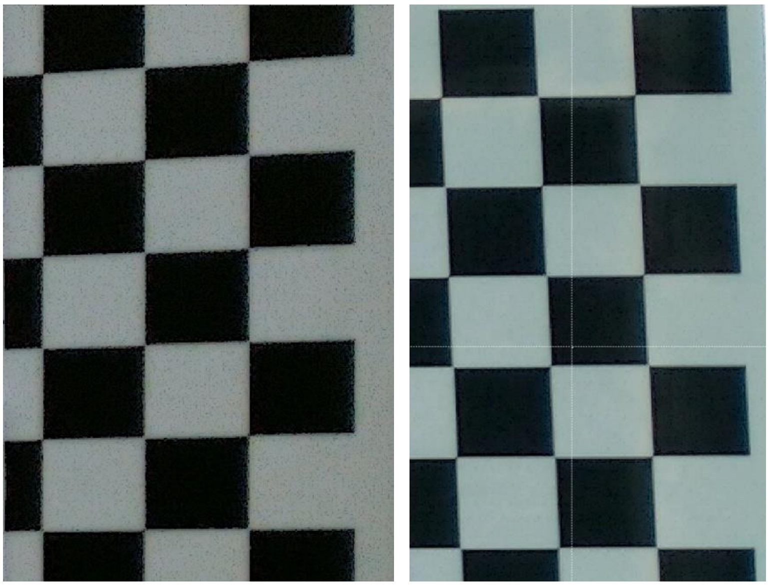


2. The transition zone in the middle of the diagonal checkerboard is too large for the naked eye to find the corners; (Left: the transition area is too large; Right: transition area is small)

3. Checkerboard black and white square edge sharpening too large white edge, resulting in less accurate corner detection; (Left image: sharpened white edge is too large; Right: sharpened white edge small)



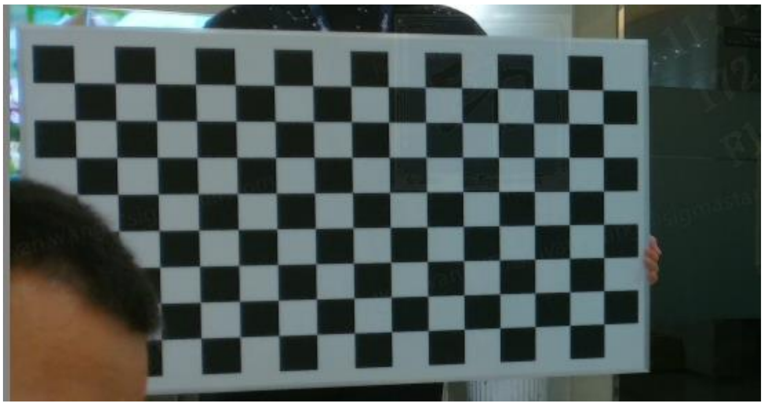
4. The picture has many noise points and high noise intensity; (Left: before dimming; Right: Increase screen brightness and increase exposure time)



5. The checkerboard calibration board should not have obvious reflection/shadow and other places that lead to uneven brightness and dark, otherwise it will lead to undetectable corners; (The following picture cannot detect the corner due to obvious reflection)

6. The black and white squares of each checkerboard calibration board must be complete in the picture, otherwise the corners will not be detected; (the black squares above the checkerboard in the following picture are incomplete, resulting in undetectable corners)

7. During the shooting, do not touch the black and white squares of the checkerboard in the calibration board, and do not block the black and white squares of the checkerboard, otherwise the corners will not be detected;



8. Relatively perfect IQ renderings (corner position is clearly visible to the naked eye) :